

Rethinking Improper Movement: Evidence from Locality Asymmetries in Mandarin Chinese

Abstract

The longstanding ban on improper movement has been derived either from domain opacity (e.g., Williams 2003) or from operation ordering (e.g., Abels 2007). This paper adjudicates between these two approaches by investigating a sharp locality asymmetry in the Mandarin *lian... dou* construction: movement to a TP-internal position can span TPs but not CPs, whereas movement to a TP-external position can span both. Crucially, the “blocked” TP-internal position is still required as an intermediate step in long-distance movement that terminates at the TP-external position. This pattern is shown to follow naturally under an operation-ordering framework, rather than a domain-opacity approach, since only the former permits long-distance movement to proceed through clause-internal intermediate positions. Building on Abels’ (2007; 2012b) feature hierarchy and phase-based model of cyclicity, this paper reformulates his Generalized Prohibition against Improper Movement as a Generalized Ban on Improper Feature Interpretation. The revised constraint applies exclusively to sequences of feature interpretation steps along the movement path, while exempting intermediate movement steps motivated by successive cyclicity. This analysis not only has repercussions for the proper analytical treatment of improper movement but also reveals a broader typological interaction between successive-cyclic movement and clause-internal operations.

Keywords: Improper movement, Williams Cycle, Cyclicity, Phases, Mandarin Chinese

1 Introduction

Locality mismatches arise when a structure permits one type of movement to proceed out of it, while simultaneously prohibiting another type. The most well-known case is illegitimate hyperraising in English, whereby $\bar{A}$ -movement can cross CP boundaries (1), whereas A-movement cannot (2). The classical account attributes this movement asymmetry to a conspiracy of two constraints: (i) a locality principle requires movement out of a CP to proceed through its specifier, an $\bar{A}$ -position (Chomsky 1973, 1981, 1986); and (ii) a ban on “improper movement”, according to which movement cannot proceed from an $\bar{A}$ -position to an A-position. (Chomsky 1973, 1981; May 1979).

- (1) Who₁ does it seem [_{CP} t₁ has left]?

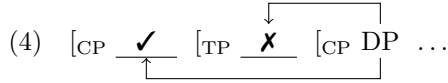
- (2) * Who₁/John₁ seems [_{CP} t₁ has left](?)

A growing body of work shows that locality mismatches extend beyond the binary distinction between A- and $\bar{A}$ -movement (e.g., Williams 1974, 2003, 2011, 2013; Müller and Sternefeld 1993; Grewendorf 2003; Abels 2007, 2012b; Neeleman and van de Koot 2010; Müller 2014a,b; Keine 2019, 2020; Poole 2023; Meadows 2024). These results motivate revising the theory of improper movement, which must be generalized to capture the distinct locality profiles of different movement types.¹

Two main approaches have been proposed for such a general theory (for discussion of other approaches, see Keine 2020). The first approach stems from the prominent work by Williams (1974, 2003, 2011, 2013) and van Riemsdijk and Williams (1981), known as the Williams Cycle. Williams (2003) formulates the Williams Cycle as the *Generalized Ban on Improper Movement*, stated in (3).

- (3) Generalized Ban on Improper Movement (GBOIM) (Poole 2023, p. 348)
 Movement to [_{Spec}, XP] cannot proceed from [_{Spec}, YP] or across YP, where Y is higher than X ($Y \succ X$) in the functional sequence (*fseq*).

The Williams Cycle is a domain-based account of improper movement. The core idea is that once movement leaves a particular domain in an embedded clause, subsequent movement must target a domain of the same or a higher type, not a lower one. In other words, it essentially renders one and the same domain opaque to some movement types but transparent to others. For example, it explains the impossibility of hyperraising in (2) as a ban on extraction from inside a CP to [_{Spec}, TP]. As schematized in (4), the entire CP domain is inaccessible for movement to TP, but not for movement to CP, because T is structurally lower C in *fseq*. For ease of exposition, I refer to this approach as DOMAIN OPACITY. Domain-opacity analyses that are more compatible with the current minimalist theory have been developed by Keine (2016, 2019, 2020), Poole (2023) and Meadows (2024).



The second approach derives locality mismatches from constraints on permissible sequences of movement operations, which I refer to as OPERATION ORDERING (Grewendorf 2003; Abels 2007, 2012b; Neeleman and van de Koot 2010). Under this approach, the relevant constraint refers to the moving element: it restricts the sequence of operations that apply to that element, rather than rendering a particular domain selectively opaque. Abels (2007, 2012b) implements this approach by positing a transitive, asymmetric, and possibly total order on movement-type-specific features, as in (5). A direct consequence of this ordering is that certain sequences of movement operations are impossible, as formalized in (6). To see how this system rules out hyperraising in (2),

¹Locality mismatches arises not only in the movement path of an individual item, but also in cases of sub-extraction and remnant movement (Müller and Sternefeld 1993; Abels 2007, 2012b; Neeleman and van de Koot 2010), as well as non-movement dependencies, such as long-distance agreement (Keine 2016, 2019, 2020) and dependent case assignment (Poole 2023). In this paper, I focus on the movement path of a single element and leave extensions to related configurations for future work. For a brief discussion, see Sect. 7.

let's assume that, in English, T bears $[\phi]$ features and C bears $[\bar{A}]$ features. Extraction from inside a CP must proceed through the CP edge, and movement to that edge can be triggered only by an $[\bar{A}]$ feature. However, under GenPIM, $[\phi]$ -triggered movement cannot apply to an element that has undergone $\bar{A}$ -movement.

- (5) The Universal Constraint on the Ordering of Operations in Language (UCOOL) (adapted from Abels 2007, p. 60)
Movement operations are hierarchically ordered, with features to the left of $\ll$ ranked lower than features to the right: $[\theta] \ll [\phi] \ll [\bar{A}]^2$
- (6) Generalized Prohibition against Improper Movement (GenPIM) (adapted from Abels 2007, p. 63)
No constituent may undergo movement triggered by $[\sigma]$ if it has undergone movement triggered by $[\tau]$, where $[\sigma] \ll [\tau]$ under UCOOL.

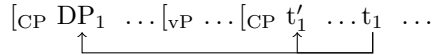
Although these two approaches are conceptually distinct, they are mostly indistinguishable on the basis of hyperraising data such as (2). Therefore, it is necessary to look elsewhere to empirically adjudicate between them. This paper takes up that task and arrives at two main conclusions.

First, I distinguish the two approaches and argue for operation ordering by examining an empirical domain where they make distinct predictions: successive cyclicity. The two approaches diverge in whether they allow clause-internal intermediate landing sites. As depicted in (7), operation ordering is compatible with the standard assumption that CPs and vPs are phases, and consequently, long-distance movement proceeds through [Spec, CP] (t'_1) and [Spec, vP] (t''_1) (Chomsky 2000, 2001, 2008). In contrast, the distribution of intermediate landing sites is more restricted under domain opacity. Under this approach, movement from [Spec, CP] to [Spec, vP] is prohibited, because movement cannot proceed from a higher projection to a lower one. As a result, domain-opacity analyses deny the phasehood of vPs and derive long-distance movement chains with [Spec, CP]-to-[Spec, CP] movement, as in (8).

- (7) Long-distance movement under operation ordering:



- (8) Long-distance movement under the domain-opacity approach:



Using evidence from (im)proper movement in the *lian... dou* construction in Mandarin Chinese (MC), I show that the predictions of operation ordering are borne out. In the *lian... dou* construction, the focus particle *lian* combines with a focus expression

² $[\bar{A}]$ represents a class of specific $[\bar{A}]$ features, such as [Top], [Foc], and [Wh]. For simplicity, I omit the internal hierarchical order of these $[\bar{A}]$ features, as the current discussion does not rely on it. The reader may refer to Abels (2007, 2012a) for a finer-grained feature hierarchy. In Sect. 4, where the relative order of $[\bar{A}]$ features becomes crucial, I will show that in MC, [Foc] ranks lower than [Top].

to form a constituent. The resulting *lian*-phrase (*lian*-XP) must move to a position above *dou*, surfacing either below or above the subject, as shown in (9). Throughout the paper, I use *TP-internal movement* and *TP-external movement* as convenient descriptive labels for these two movement types.

- (9) <lian chongzi₁ > Zhangsan <lian chongzi₁ > dou gan chi t₁.
 lian insect Zhangsan lian insect dou dare eat
 ‘Zhangsan dares to eat even insects.’

As schematically shown in (10a–b), TP-external movement can cross TPs and CPs boundaries.³ On the basis of reconstruction effects and the strict local dependency between *dou* and the fronted *lian*-phrase (see Sect. 2), I argue that TP-external movement passes through a TP-internal position, as indicated by t'₁. Crucially, the domain hosting this intermediate landing site is structurally lower in *fseq* than the domain from which the focus moves (TP in (10a) and CP in (10b)). Under domain opacity, however, such a higher-to-lower step is predicted to be impossible. Given that domain opacity faces an empirical problem that does not arise under operation ordering, I take this as evidence in favor of operation ordering.

- (10) (Im)proper movement in the *lian*...*dou* construction
 a. lian-XP₁ Subject t'₁ dou ... [TP ... t₁ ...
 b. lian-XP₁ Subject t'₁ dou ... [CP ... [TP ... t₁ ...
 c. Subject lian-XP₁ dou ... [TP ... t₁ ...
 d. * Subject lian-XP₁ dou ... [CP ... [TP ... t₁ ...

Second, I show that, to explain the data properly under an operation-ordering analysis, the domain of application of the constraint on improper movement must be refined. In existing operation-ordering analyses, the constraint against improper movement applies to all movement steps. I argue that *not all* movement steps are subject to this constraint, a view previously advanced by Müller (2014a) and Meadows (2024). This claim is supported by a crucial locality asymmetry in (10). I show that TP-internal movement can proceed out of TPs (10c) but not CPs (10d), even when the focus undergoes $\bar{A}$ -movement. A closer comparison between grammatical TP-external movement (10b) and ungrammatical TP-internal movement (10d) reveals that the derivation of the former properly contains that of the latter. Specifically, although the TP-internal position is “blocked” as a final landing site in (10d), it serves as an

³In MC, complement clauses differ in their structural sizes. Yan and Meadows (2025) show that clauses selected by *think*-type verbs permit topicalization within their domain, and the topicalized object can precede sentence-level adverbs like *xingkuai* ‘fortunately’, as in (i). In contrast, clauses selected by control verbs do not allow both, as in (ii). For concreteness, I take the former to be CPs, and the latter to be (defective) TPs. The syntactic properties of these clauses, such as finiteness, are orthogonal to the present discussion; see Huang (2024) and references therein.

(i) Wassily renwei [CP [na-ben xiaoshuo-a]₁ xingkuai [TP Piet kan-le san-bian t₁]]
 Wassily think that-CLF novel-TOP fortunately Piet read-ASP three-time
 ‘Wassily thinks that, that novel, fortunately, Piet read three times.’

(ii) *Wassily qiangpo Piet [TP [na-ben xiaoshuo-a]₁ xingkuai kan-le san-bian t₁]
 Wassily force Piet that-CLF novel-TOP fortunately read-ASP three-time
 Intended: ‘As for that novel, fortunately, Wassily forced Piet to read three times.’

intermediate landing site in (10b). This contrast motivates a key generalization that any analysis of locality mismatches must capture: a position that imposes locality constraints on terminal movement into it does not impose the same constraints on intermediate movement into it. Building on Abels’ (2007; 2012b) analysis, I reformulate GenPIM as a constraint that applies selectively to terminal movement while ignoring intermediate movement. This refinement entails that *some but not all* movement steps are relevant to determining whether a movement chain is improper.

The rest of the paper is organized as follows. In Sect. 2, I present the core data underlying the pattern schematized in (10). On the basis of these data, Sect. 3 develops an operation-ordering analysis. Sect. 4 shows that the proposed analysis correctly predicts the pattern of cross-clausal movement in the *lian...dou* construction. Sect. 5 shows that the pattern in (10) is not an isolated case. Instead, it instantiates a particular type of interaction between syntactic operations across languages. In Sect. 6, we zoom out to consider an intriguing typological generalization regarding the connection between the locality of a movement type (short vs. long) and the structural height of that movement type’s landing site (low vs. high). Sect. 7 concludes.

2 Cross-clausal movement in the *lian...dou* construction

2.1 TP-internal movement

In MC, the *lian...dou* construction involves focus movement and yields an interpretation comparable to English *even* (e.g., Qu 1994; Shyu 1995; Badan 2008; Chen, F. 2023; Cao 2025a,b). As shown in example (11), *dou* occupies a preverbal position. The focus particle *lian* combines with the focused object to form a *lian*-phrase. The resulting *lian*-phrase moves to a TP-internal position between the subject and *dou*.

- (11) Zhangsan *lian* *chongzi*₁ *dou* *gan* *chi* *t*₁.
 Zhangsan *lian* insect *dou* dare eat
 ‘Zhangsan dares to eat even insects.’

TP-internal movement of DPs exhibits typical A-properties (Shyu 1995; Chen 2023; Cao 2025a,b). For example, movement of DP foci creates new antecedents for anaphors at the landing site, as illustrated by the contrast between (12) and (13).

- (12) *Wo *bi* *ta-ziji*₁ *de* *erzi* [_{TP} *chumai* *le* *Lisi*₁].
 I force s/he-self POSS son betray ASP Lisi
 Intended: ‘I forced his₁ own son to betray Lisi₁.’
- (13) Wo *lian* *Lisi*₁ *dou* *bi* *ta-ziji*₁ *de* *erzi* [_{TP} *chumai* *le* *t*₁].
 I *lian* Lisi *dou* force s/he-self POSS son betray ASP
 ‘Even Lisi₁, I forced his₁ own son to betray him₁.’

Another A-property that will become important is that moved DPs do not reconstruct below *dou*.⁴ Example (14) is ungrammatical because the anaphor cannot be

⁴Reconstruction effects in A-movement vary across languages. In English, A-movement reconstructs for scope and variable binding, but only in limited environments (e.g., Fox 1999; Lebeaux 2009). Lechner (1998)

bound under reconstruction. Scope reconstruction is not possible either. Example (15) allows only the surface scope reading, indicating that the fronted DP does not reconstruct into the scope of the universal *meige xuesheng*.

- (14) *Wo lian [_{DP} ta-ziji₂ de fangzi]₁ dou bi Li₂ [_{TP} maidiao le t₁].
 I lian s/he-self POSS house dou force Li sell ASP
 Intended: ‘I forced Li₂ to sell even his₂ own house.’
- (15) Li lian [yi-qian de jiang-jin]₁ dou fengei le meige xuesheng t₁.
 Li lian one-thousand ADJ bonus-cash dou give ASP every student
 $\exists > \forall$: ‘Lisi distributed even one thousand yuan (in total) to every student.’
 $*\forall > \exists$: ‘Zhangsan gave each student even one thousand yuan.’

Movement in the *lian...dou* construction may also target VP foci. In sharp contrast to DP foci, fronted VPs reconstruct for anaphor binding (16) and scope (17). Since reconstruction for binding and scope is a typical $\bar{A}$ -property and VPs generally do not undergo A-movement (e.g., Barss 1986; Huang 1993; Heycock 1995), I follow Cao (2025a,b) in taking this contrast as evidence that VP foci undergo $\bar{A}$ -movement.

- (16) Wo lian [_{VP} maidiao ta-ziji₂ de fangzi]₁ dou bi Li₂ [_{TP} zuo le t₁].
 I lian sell s/he-self POSS house dou force Li do ASP
 ‘I even forced Li₂ to sell his₂ own house.’
- (17) Li₁ lian [_{VP} gei ta₁ yi-qian de hong-bao]₂ dou bi meige
 Li lian give him₁ one-thousand ADJ red-envelope dou force every
 xuesheng [_{TP} zuo guo t₂].
 student do EXP
 $\forall > \exists$: ‘Li₁ forced each student even to give him₁ one thousand yuan.’
 $*\exists > \forall$: ‘Li₁ forced every child even to give one thousand yuan (in total).’

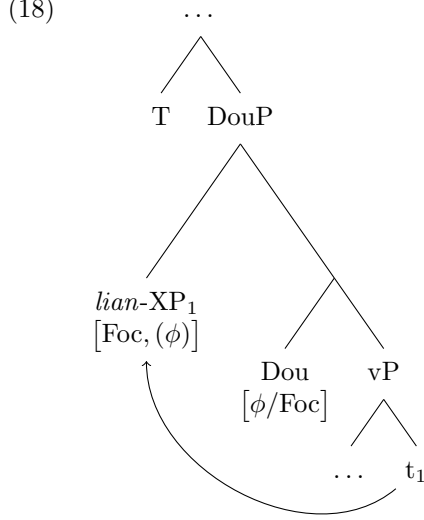
To capture this state of affairs, I assume that A- and $\bar{A}$ -properties of movement follow from properties of the agreeing feature(s) present on the probing head (Chomsky 2007; Obata 2010; Abels 2012a,b; van Urk 2015). Specifically, I adopt the analyses proposed by Chen (2023) and Cao (2025a). The head *dou* bears a composite [ϕ /Foc(us)] probe and triggers movement of *lian*-phrases to its specifier, as depicted in (18).⁵

shows that superlocal scrambling in German reconstructs for scope but not for variable binding. Cases of A-movement with no reconstruction effects are also attested. In MC, DP fronting to a preverbal position generally does not reconstruct for binding and scope, as shown in (i) and (ii). Another example is superlocal A-scrambling in Hindi—i.e., scrambling a direct object over an indirect object while remaining below the subject (Poole and Keine 2024). The cross-linguistic variation in reconstruction under A-movement is a separate issue (for further discussion, see Poole and Keine 2024). Nevertheless, the overarching A/ $\bar{A}$ distinction in reconstruction remains valid in MC: reconstruction under A-movement is more limited than under $\bar{A}$ -movement. As I will show later, $\bar{A}$ -movement in MC does show reconstruction effects.

- (i) (Shyu 1995, p. 104)
 *Wo [naxie ta-ziji₁ de shu]₂ yijing jiao Zhangsan₁ xian na-zou le t₂
 I those s/he-self POSS book already ask Zhangsan first take-away ASP
 ‘I have asked Zhangsan₁ to take away his₁ own books.’
- (ii) Li yuanyi [yi-qian-kuai de jiang-jin]₁ fengei meige xuesheng t₁.
 Li willing one-thousand-yuan ADJ bonus-cash give every student
 $\exists > \forall$: ‘Li is willing to distribute one thousand yuan (in total) to every student.’
 $*\forall > \exists$: ‘Li is willing to give each student even one thousand yuan.’

⁵The idea of a TP-internal focus projection goes back to Ernst and Wang (1995) and Shyu (1995).

Under this model, the contrast between A-movement of DP foci and $\bar{A}$ -movement of VP foci follows from a feature-matching asymmetry in the probing process. DP foci match the $[\phi/\text{Foc}]$ probe in both $[\phi]$ and $[\text{Foc}]$ features. The former determines that the resulting movement chain exhibits A-properties, and the latter ensures that movement yields a focus interpretation. By contrast, VP foci agree with the $[\phi/\text{Foc}]$ probe only in the $[\text{Foc}]$ feature, resulting in $\bar{A}$ -movement of these foci.⁶



With respect to locality, TP-internal movement of all categories can proceed out of TPs but not out of CPs. The ability of DP and VP foci to move out of TPs has been illustrated in examples (13) and (16–17). Examples (19) and (20) show that neither a DP focus nor a VP focus can move from the embedded CP to $[\text{Spec}, \text{DouP}]$ in the matrix clause. This indicates that the locality constraints on TP-internal movement must be dissociated from the A/ $\bar{A}$ -properties of the movement chain, but are sensitive to the category of the clause from which the focus moves.

- (19) * Lisi lian laobao₁ dou juede [_{CP} Zhangsan ma guo t₁]
 Lisi lian boss dou think Zhangsan scold EXP
 Intended: ‘Even the boss₁, Lisi thinks Zhangsan scolded him₁.’
- (20) * Lisi lian [_{VP} ma laobao]₁ dou juede [_{CP} Zhangsan gan zuo t₁].
 Lisi lian scold boss dou think Zhangsan dare do
 Intended: ‘Even scold the boss, Lisi thinks that Zhangsan dares to do.’

2.2 TP-external movement

TP-external movement involves subsequent topicalization targeting TP-internal *lian*-phrases (Shyu 1995; Badan 2008; Badan and Del Gobbo 2015; among others). Evidence

⁶I assume that, in cases of movement of VP foci, the $[\phi]$ probe is either deleted or assigned a default value in the sense of Preminger (2009, 2011, 2014). For discussion of how the composite probe triggers A-movement of DP foci but $\bar{A}$ -movement of VP foci, see Sect. 3 and Cao (2025a).

for the topichood of TP-external *lian*-DPs comes from the contrast between examples (22–23). Like typical topicalization of DPs such as example (21), TP-external *lian*-DPs can co-occur with the topic marker *a*, as in (22), whereas TP-internal *lian*-DPs cannot, as in (23).⁷

- (21) Lisi(-a)₁ Zhangsan hen bu xihuan t₁.
 Lisi(-TOP) Zhangsan very NEG like
 ‘Lisi, Zhangsan does not like (him/her),’
- (22) Lian Lisi(-a)₁ Zhangsan t’₁ dou bu xihuan t₁.
 lian Lisi(-TOP) Zhangsan dou NEG like
 ‘Even Lisi, Zhangsan does not like (him/her).’
- (23) Zhangsan lian Lisi(*-a)₁ dou bu xihuan t₁.
 Zhangsan lian Lisi(*-TOP) dou NEG like
 ‘Zhangsan does not trust even Lisi.’

Although TP-external *lian*-VPs cannot be marked by the topic marker *a*, I argue that they are nevertheless derived by topicalization, on a par with TP-external *lian*-DPs. First, VPs can be topicalized to the left periphery, as in (24). Second, TP-external movement of VPs, as in (25), resembles VP topicalization (24) in that the fronted VP must reconstruct to its base position. Since topicalized VPs and TP-external *lian*-VPs move to the same domain and exhibit the same $\bar{A}$ -properties, the most parsimonious analysis is that TP-external movement of VP foci is driven by topicalization.

- (24) (Huang 1993, p. 119)
 [VP piping ta-ziji*_{1/2}]₃ Zhang₁ zhidao [CP Li₂ juehui bu hui t₃]
 criticise s/he-self Zhang know Li definitely not will
 ‘Criticize himself*_{1/2}, Zhang₁ knows that Li₂ definitely will not.’
- (25) Lian [VP ma ta-ziji*_{1/2} de pengyou]₃ Zhang₁ t’₃ dou juede [CP Li₂ zuo
 lian scold s/he-self POSS friend Zhang dou think Li do
 guo t₃].
 EXP
 ‘Even scold his*_{1/2} own friends, Zhang₁ thinks Li₂ did.’

TP-external movement of all categories is essentially unbounded. It can extract DP and VP foci from TPs (26–27) and CPs (28–29). The full pattern of cross-clausal movement in the *lian...dou* construction is summarized in (30).

- (26) Lian [DP zhe-ben shu]₁ wo dou bi Zhang [TP kan-wan le t₁].
 lian this-CLF book I dou force Zhang read-finish ASP
 ‘I forced Zhang to read even this book.’

⁷In English, foci associated with *even* cannot be topicalized using the “as for” construction, as shown in (i). I contend that example (i) is unacceptable because the particle *even* fails to take propositional scope. First, the particle *even* cannot take scope over the comment clause because it is embedded within the “as for” phrase. Second, since “as for” introduces a base-generated topic, the embedded *even* cannot reconstruct to a scope position within the comment clause. In contrast, as we will see below, TP-external *lian*-phrases reconstruct to the position where the focus is interpreted.

(i) *As for even Lisi, Zhangsan does not like (him/her).

- (27) Lian [_{VP} kan-wan zhe-ben shu]₁ wo dou bi Zhang [_{TP} zuo le t₁].
 lian read-finish this-CLF book I dou force Zhang do ASP
 ‘I forced Zhang to even read this book.’
- (28) Lian laobao₁ Lisi dou jue de [_{CP} Zhangsan ma guo t₁]
 lian boss Lisi dou think Zhangsan scold EXP
 ‘Even the boss₁, Lisi thinks Zhangsan scolded him₁.’
- (29) lian [_{VP} ma laobao]₁ Lisi dou jue de [_{CP} Zhangsan zuo guo t₁]
 lian scold boss Lisi dou think Zhangsan do EXP
 ‘Even scold the boss, Lisi thinks that Zhangsan dares to do.’
- (30) (Im)proper movement in the *lian...dou* construction
- a. lian-XP₁ Subject t'₁ dou ... [_{TP} ... t₁ ...
 - b. lian-XP₁ Subject t'₁ dou ... [_{CP} ... [_{TP} ... t₁ ...
 - c. Subject lian-XP₁ dou ... [_{TP} ... t₁ ...
 - d. * Subject lian-XP₁ dou ... [_{CP} ... [_{TP} ... t₁ ...

2.3 Movement out of doubly embedded structures

A crucial observation from previous sections is that, in the *lian...dou* construction, there is a systematic connection between the structural height of a movement type’s landing site and the range of clauses from which that movement type may proceed. Namely, the higher the landing site of a movement type, the broader the range of structures it can span.

Further evidence for this connection comes from movement out of doubly embedded structures. Example (31) establishes the baseline configuration, in which a CP is embedded within a TP, which in turn is embedded in a matrix clause. Example (32) shows that the object of the most deeply embedded CP cannot move to a position below the matrix subject; only a TP-external position is available. Example (33) differs from (32) in that the fronted DP originates within a TP. In this case, movement can target a TP-internal position in the matrix clause. Crucially, (32) and (33) differ only in the category of the clause containing the base position of the moved DP: CP in (32) versus TP in (33). This suggests that TP-internal movement cannot extract a focus from inside a CP, regardless of how deeply that CP is embedded.⁸

- (31) Li₁ dasuan [_{TP} chengren [_{CP} ta₁ ma guo laoshi]].
 Li plan admit s/he scold EXP teacher
 ‘Li₁ planned to admit that he₁ scolded the teacher.’
- (32) <Lian laoshi₂ > Li₁ <*lian laoshi₂ > dou dasuan [_{TP} chengren [_{CP} ta₁ ma
 lian teacher Li lian teacher dou plan admit s/he scold
 guo t₂]].
 EXP
 ‘Even the teacher₂, Li₁ planned to admit that he₁ scold him₂.’

⁸Movement of VP foci exhibits the same pattern in doubly embedded structures; the relevant examples are omitted from the main text due to space limitations.

- (33) <Lian laoshi₂ > Li₁ <Lian laoshi₂ > dou dasuan [TP jixu [TP ma t₂]].
lian teacher Li lian teacher dou plan continue scold
‘Even the teacher₂, Li₁ planned to continue scolding him₂.’

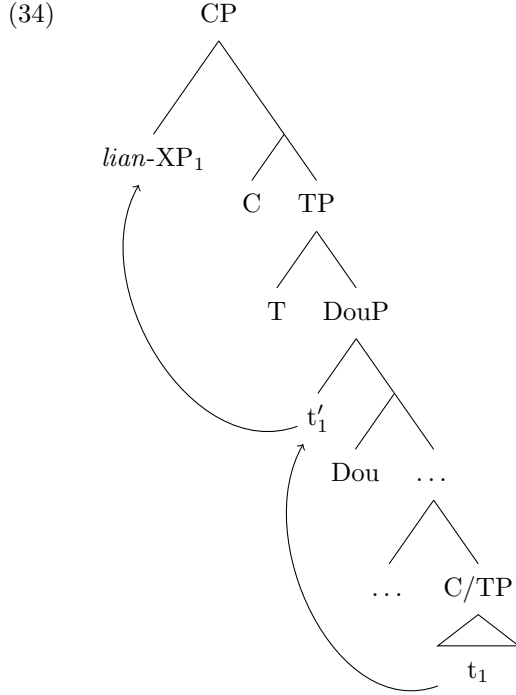
2.4 Intermediate movement in the matrix clause

In TP-external movement, the *lian*-phrase consistently moves through [Spec, DouP] in the matrix clause, as indicated by t'₁ in (34). For the sake of readability, I omit intermediate landing sites induced by phase impenetrability (i.e., [Spec, vP] and [Spec, CP]), though their presence is assumed. A full derivation that includes these intermediate landing sites is given in Sect. 4.

The first piece of evidence concerns the local dependency between *dou* and the *lian*-phrase. In TP-internal movement, the fronted *lian*-phrase must immediately c-command *dou* (Zhang 1997; Cao 2025b).⁹ I hereafter refer to this constraint as the *Immediate C-command Condition*. This condition rules out examples of DP fronting in (35) and VP fronting in (36), in which the adverb *jingchang* ‘often’ may attach either above the fronted *lian*-phrase or below *dou*, but not between them. Recall that I adopt a composite-probe analysis to TP-internal movement. The immediate c-command relation between *dou* and the fronted *lian*-phrase follows straightforwardly under the standard assumption that feature valuation between a probe and its goal depends on a specifier–head relation.¹⁰

⁹A constituent α immediately c-commands a constituent β iff there is no constituent γ such that α asymmetrically c-commands γ and γ asymmetrically c-commands β . α asymmetrically c-commands β iff α c-commands β and β does not c-command α .

¹⁰The reader may refer to Cao (2025b) for a semantic-based account.



- (35) Mali <jingchang> lian fan₁ <*jingchang> dou <jingchang> bu chi t₁.
 Mary often lian meal often dou often NEG eat
 ‘Mary often does not even eat.’
- (36) Wu <jingchang> lian [_{VP} chi sheng-fan]₁ <*jingchang> dou <jingchang>
 Wu often lian eat left-food often dou often
 bi Lisi duo t₁.
 force Lisi do
 ‘Wu often forces Lisi even to eat leftovers.’

In examples (37) and (38), the TP-external *lian*-phrase is not linearly adjacent to *dou*. However, I argue that these are not true counterexamples. Given that *dou* is also obligatory in TP-external movement and that the same focus interpretation is obtained, the same features must be valued in the same specifier–head configuration. Otherwise, the derivation would crash, as the features on *dou* would remain unvalued. Therefore, in order to satisfy the immediate c-command condition, the moving *lian*-phrase must have occupied [Spec, DouP] at some stage of the derivation. The only difference is that, in cases of TP-external movement, [Spec, DouP] serves as an intermediate landing site, as indicated by t₁ in (37–38).

- (37) lian xizhuren₁ Li t₁ dou ma guo t₁.
 lian dean Li dou scold EXP
 ‘Li even scolded the dean.’

- (38) *lian* [_{VP} *chi guoqi-de shiwu*]₁ *Li* *t'*₁ *dou bi* *Wu* *zuo guo* *t*₁.
lian eat expired-ADJ food *Li* *dou* force *Wu* do EXP
 ‘Li forced Wu to even eat the expired food.’

Further support comes from reconstruction effects. As with TP-internal movement, TP-external movement of DPs does not reconstruct below *dou*. However, TP-external *lian*-DPs do exhibit reconstruction effects in the domain above *dou*. In example (39), the anaphor *ta-ziji* can only be bound by the subject *Zhang*, not by the lower argument *Li*. I assume that the binding relation is determined precisely at [Spec, DouP] (*t'*₃ in (39)), where the features on the head *dou* are valued.

- (39) *Lian* [_{DP} *ta-ziji*_{1/*2} *de pengyou*]₃ *Zhang*₁ *t'*₃ *dou bi* *Li*₂ [_{TP} *ma guo* *t*₃].
lian s/he-self POSS friend *Zhang* *dou* force *Li* scold EXP
 ‘Even his_{1/*2} own friends, Zhang₁ forced Li₂ to scold (them).’

Long-distance TP-external movement of DP foci also reconstruct to the domain above *dou*. In examples (40) and (41), the binding possibilities of the anaphor track the position of *dou*, as highlighted by boldface.¹¹ In example (40), where *dou* occurs in the matrix clause, the anaphor can only be bound by the matrix subject.¹² By contrast, when *dou* appears in the embedded CP in (41), the anaphor can be bound by either the matrix subject or the embedded subject. This contrast is best explained if the moved DP reconstructs to [Spec, DouP], but not to any position below it.

- (40) *Lian* [_{DP} *ta-ziji*_{1/*2} *de pengyou*]₃ *Wu*₁ *t'*₃ ***dou*** *juede* [_{CP} *Li*₂ *ma guo* *t*₃].
lian s/he-self POSS friend *Wu* *dou* think *Li* scold EXP
 ‘Even his_{1/*2} own friends, Wu₁ thinks Li₂ scolded (them).’
- (41) *Lian* [_{DP} *ta-ziji*_{1/2} *de pengyou*]₃ *Wu*₁ *juede* [_{CP} *Li*₂ *t'*₃ ***dou*** *ma guo* *t*₃].
lian s/he-self POSS friend *Wu* think *Li* *dou* scold EXP
 ‘Even his_{1/2} own friends, Wu₁ thinks Li₂ scolded (them).’

¹¹The position of *dou* correlates with the scope of the focus interpretation (Shyu 1995, 2018). In the examples below, the matrix predicate is semantically negative and licenses a downward-entailing context. When *dou* sits in the embedded CP as in (i), the focus is interpreted in the scope of this negative predicate. By contrast, when *dou* appears in the matrix clause, the focus receives a wide-scope interpretation, as in (ii). The interaction between downward-entailing licensors and *even*-like particles is an independent topic; see Crnič (2011) and references therein.

- (i) *Lian* [_{jufa} *jiegou*]₁ *wo* *hen-nan* *xiangxin* [_{CP} *Lisi* ***dou*** *du-de-dong* *t*₁].
lian [syntactic structure] I very-hard believe *Lisi* *dou* read-able-understand
 ‘It is hard for me to believe that Lisi understand even *Syntactic Structure*.’
 Focus interpretation: *Syntactic Structure* is the least likely thing for Lisi to understand. (i.e. *Syntactic Structure* is difficult for Lisi to understand.)
- (ii) *Lian* [_{jufa} *jiegou*]₁ *wo* ***dou*** *hen-nan* *xiangxin* [_{CP} *Lisi* *du-de-dong* *t*₁].
lian [syntactic structure] I *dou* very-hard believe *Lisi* read-able-understand
 ‘It is hard for me to believe that Lisi understand even *Syntactic Structure*.’
 Focus interpretation: *Syntactic Structure* is the least likely thing that it is hard for me to believe that Lisi understands. (i.e. *Syntactic Structure* is easy for Lisi to understand.)

¹²One may wonder why reconstruction below *dou* is blocked. I assume that this follows from the properties of the features valued at this position. In cases of *lian*-DPs, the valued [ϕ] feature prevents reconstruction below *dou*. By contrast, VP foci agree with the [ϕ /Foc] probe only in the [Foc] feature. As shown in example (24), TP-external *lian*-VPs exhibit reconstruction to their base positions.

In a nutshell, TP-external movement necessarily passes through [Spec, DouP] in the matrix clause. Crucially, this intermediate landing site is not a byproduct of clause-internal phases (e.g., vP). Rather, it is required by the feature valuation between *dou* and the moving *lian*-phrase.

The existence of this “special” intermediate landing site has important implications for accounts of locality mismatches. First, it rules out analyses based on domain opacity (e.g., Williams 2003; Keine 2019; Poole 2023). These analyses wrongly predict that, once the moving element reaches the embedded [Spec, CP], the next landing site must be the matrix [Spec, CP], skipping all positions between them.

Second, it reveals a key property of locality mismatches between TP-internal and TP-external movements. Movement from within a CP to [Spec, DouP] is well-formed only when it is intermediate rather than terminal. This pattern entails a core empirical generalization that any adequate account of locality mismatches must capture: a position that imposes locality constraints on terminal movement into it does not impose the same constraints on intermediate movement into it. In other words, there is a fundamental asymmetry between terminal and intermediate movement with respect to locality constraints. In the following sections, I develop an account based on operation ordering and show how it accounts for the Mandarin facts.

3 The proposal

Building on Abels’ (2012b) local derivational approach to movement, I develop a movement system that distributes labor across two components. First, the system incorporates a feature-based theory of cyclicity and phases (see also Chomsky 1995, 2007, 2008; Georgi 2014; van Urk 2015). Under this view, the locality of a movement type follows from the distribution of movement-type-specific features across phase heads (see also Rizzi 1997, 2004; Legate 2011; Obata and Epstein 2011; Abels 2012a,b; van Urk 2015). Second, it posits a hierarchy of features and their associated operations (see also Müller and Sternefeld 1993; Grewendorf 2003; Neeleman and van de Koot 2010). I follow Abels (2012b) and van Urk (2015) in distinguishing terminal from intermediate movement in terms of feature interpretability. While both terminal and intermediate movement steps are driven by feature valuation, only terminal movement triggers feature interpretation. For expository purposes, terminal movement is referred to as *feature interpretation* steps, and intermediate movement as *feature valuation* steps. The idea that locality constraints on movement selectively apply terminal movement is formulated a constraint on admissible sequences of feature interpretation steps along a movement path. A movement chain is deemed improper if a movement step that interprets a feature $[\tau]$ is followed by a movement step that interprets a feature $[\sigma]$, with $[\tau]$ ranking higher than $[\sigma]$ in the hierarchy.

3.1 Successive cyclic movement

I assume a derivational model of syntax in which syntactic structure is built bottom-up via applications of Merge and Agree, and only local domains of the structure are accessible at any point in the derivation (Chomsky 1993, 1995, 2001).

Features involved in Agree are specified for their morphosyntactic and interpretive properties and are classified into [A] features (e.g., [D] and [ϕ]) and [$\bar{A}$] features (e.g., [Wh] and [Topic]) (Rizzi 1997; Abels 2012a,b; van Urk 2015). These features can be either valued ([F]) or unvalued ([uF]). Unvalued features must acquire a value during the derivation; otherwise, they are illicit objects at the interface. I adopt the *feature-sharing* implementation of Agree proposed by Pesetsky and Torrego (2007) (see also Frampton and Gutmann 2000; Legate 2005; Bobaljik 2008). In this system, when two features are shared, valuation occurs as long as one of them is inherently valued. By adopting the feature-sharing system, we gain the technical ability to treat valuation and interpretability as independent notions. This distinction will become crucial later in distinguishing terminal from intermediate movement steps.

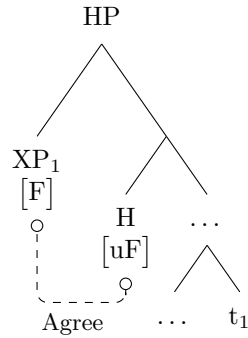
Movement is modeled as Internal Merge, that is, the combination of syntactic objects that stand in a part-whole relation. Movement does not apply freely but is triggered only as a last resort, as formalized in (42).

(42) Last Resort (Abels 2012b, p. 105):

A constituent α may only be merged—internally or externally—if that leads to the immediate valuation of a feature.

Following standard approaches to agreement, I assume that feature valuation depends on certain syntactic dependencies between a head bearing an unvalued feature (probe) and a syntactic object bearing a matching valued feature (goal). A probe that requires upward valuation (where the goal c-commands the probe) may trigger movement of the closest goal to its specifier.¹³ This process is depicted in (43), where the dashed line indicates feature sharing, and t_1 marks the base position of XP.¹⁴

(43) Internal merge (movement) of XP



That movement proceeds in a successive cyclic fashion is conceptually motivated by Chomsky (1973, 1977) and has received a wide range of evidence since then (for recent overviews, see Abels 2012b; van Urk 2020). I implement successive cyclicity

¹³In this paper, I assume that all phrasal movement results from the successive application of Merge and Agree. The main arguments are compatible with the alternative view that Agree applies before Merge. On this view, movement is triggered by probes requiring downward valuation (where the probe c-commands the goal) and simultaneously bearing a secondary structure-building property in the sense of Heck and Müller (2007). I remain agnostic about the directionality of Agree, as nothing in this paper hinges on this issue. For discussion, see Bjorkman and Zeijlstra (2019) and references therein.

¹⁴A constituent β is closer to α than γ if α asymmetrically c-commands both β and γ , and β asymmetrically c-commands γ .

with the *Phase Impenetrability Condition* (Chomsky 2000, 2001), as in (44). It requires that a goal, which is to value a probe on a head, must move through the specifiers of all phases that intervene between its base position and the specifier of that head. I assume that CPs and vPs constitute phases, and that probes may be distributed arbitrarily over phase heads. In other words, phase heads may bear any probes.

- (44) Phase Impenetrability Condition (PIC) (Chomsky 2000, p. 108)
 In a phase α with the head H, the domain of H is not accessible to operations outside α ; only H and its edge are accessible to such operations.

For illustration, consider the derivation of successive-cyclic *wh*-movement in English. For expository purposes, intermediate landing sites for movement triggered by [F] are notated as “ $t_{[F]}$ ” with primes. In (45), each step is licensed by Last Resort because it results in the valuation of a [uWh] probe. In (43b, d, f, and h), this is represented as follows. The *u* in [uWh] on C and *v* is deleted once these probes are valued, after the *wh*-phrase moves to their specifiers.

- (45) What does John think Mary will buy?
- a. $[_{v'} v_{[uWh]} [\dots \text{what}_{[Wh]} \dots]$
 - b. $[_{vP} \text{what}_{[Wh]} [_{v'} v_{[Wh]} [\dots]]$
 - c. $[_{C'} C_{[uWh]} [_{vP} \text{what}_{[Wh]} [_{vP} v_{[Wh]} \dots]]$
 - d. $[_{CP} \text{what}_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'_{[Wh]} [_{v'} v_{[Wh]} \dots]]$
 - e. $[_{v'} v_{[uWh]} [_{CP} \text{what}_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'_{[Wh]} [_{v'} v_{[Wh]} \dots]]$
 - f. $[_{vP} \text{what}_{[Wh]} [_{v'} v_{[Wh]} [_{CP} t''_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'_{[Wh]} [_{v'} v_{[Wh]} \dots]]$
 - g. $[_{C'} C_{[uWh]} [_{vP} \text{what}_{[Wh]} [_{v'} v_{[Wh]} [_{CP} t''_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'_{[Wh]} [_{v'} v_{[Wh]} \dots]]$
 - h. $[_{CP} \text{what}_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'''_{[Wh]} [_{v'} v_{[Wh]} [_{CP} t''_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'_{[Wh]} \dots]]$

Note that the system not only allows successive-cyclic movement, but also enforces it. The derivation will crash if any movement step is skipped or not licensed by the Last Resort condition. Suppose that in (45d), the *wh*-phrase moves directly from its base position to the [Spec, CP]. (46a) shows one possible derivation where the lower *v* head does not bear a [uWh] feature. This structure is ruled out by the PIC, as the *wh*-phrase is inaccessible to the [uWh] probe on C from the domain of the lower *v* head. In (46b), the lower *v* head bears a [uWh] probe. This structure does not converge for two reasons. First, just like (46a), it violates the PIC. Second, the [uWh] feature on *v* remains unvalued, resulting in an illicit object at the interface.

- (46) Violations of the Phase Impenetrability Condition
- a. $*[_{CP} \text{what}_{[Wh]} [_{C'} C_{[Wh]} [_{vP} v \dots t_{[Wh]}]]$
 - b. $*[_{CP} \text{what}_{[Wh]} [_{C'} C_{[Wh]} [_{vP} v_{[uWh]} \dots t_{[Wh]}]]$

As shown in (47), the *wh*-phrase cannot move to [Spec, vP] if the *v* head does not bear a [uWh] probe. This is because intermediate movement to [Spec, vP] violated the Last Resort condition.

- (47) Violations of Last Resort at vP
- $*[_{CP} \text{what}_{[Wh]} [_{C'} C_{[Wh]} [_{vP} t'_{[Wh]} [_{v'} v \dots t_{[Wh]}]]$

3.2 Terminal versus intermediate movement

Terminal movement is typically associated with morphosyntactic effects (e.g., ϕ -agreement and case assignment) or interpretive effects (e.g., scope-taking and information-structure properties) that are unavailable in intermediate movement steps. Therefore, virtually all approaches must distinguish terminal from intermediate movement in one way or another (for a recent overview, see van Urk 2020). In the feature-based approach I adopt here, all movement steps are triggered by designated features but are distinguished by their interpretability (Chomsky 2000, 2001; Abels 2012b; Georgi 2014; van Urk 2015).¹⁵ Specifically, although both terminal and intermediate movement are driven by feature valuation, only terminal movement feeds feature interpretation. For expository purposes, I refer to terminal movement as *feature interpretation* steps and intermediate movement as *feature valuation* steps.

Note that feature interpretation steps should not be equated with the notion of the *final landing site*.¹⁶ The *final landing site* refers to the surface position of the moved element. By contrast, feature interpretation steps are defined relative to the interpretation of specific features. A single movement chain may contain multiple feature interpretation steps, each corresponding to the interpretation of specific features along the movement path. However, the final landing site is the unique position where the moved element is pronounced. In other words, while the final landing site always involves feature interpretation, not all feature interpretation steps correspond to the final landing site. In the following, I outline Abels' (2012b) mechanism of feature interpretation. A full discussion of possible sequences of feature interpretation steps is deferred to the next section.

To facilitate the following discussion, I label the steps of successive-cyclic movement according to their triggers and landing positions, as illustrated in (48).

- (48) A movement step is defined as $M_{\beta}^{[F]}$ iff $[F]$ is the trigger for moving a syntactic object α into the specifier of a head β , with β bearing a $[uF]$ probe.
- (49) The Universal Constraint on the Ordering of Operations in Language (UCOOL) (adapted from Abels 2007, p. 60)
 Movement operations are hierarchically ordered, with a feature to the left of $\ll$ ranked lower than features to the right: $[\theta] \ll [\phi] \ll [\bar{A}]$

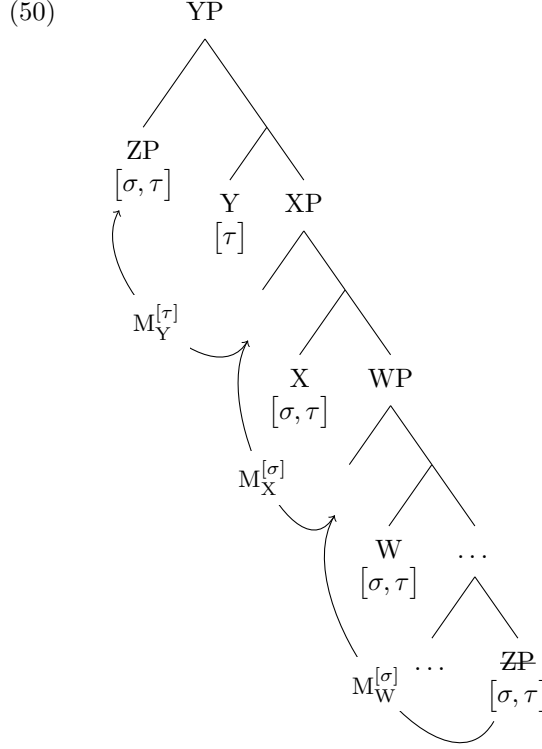
In identifying the trigger of each movement step, the core component is a universal hierarchy of features, which dictates the order in which movement operations must proceed, as shown in (49). This hierarchy stipulates that $[\theta]$ -related operations apply first, followed by operations triggered by $[\phi]$, and finally by $[\bar{A}]$ operations.

When a probing head carries one feature, and this feature is valued as the result of movement, it automatically serves as the trigger. However, there are also situations

¹⁵I assume that the interpretability of features is a relative notion. A feature may be interpretable with respect to one or both interfaces. The C-I interface (LF) can interpret features that have semantic import, such as features involving tense and $[\phi]$ features. The SM interface (PF) can interpret features that are relevant for articulation. For relevant discussion, see Pesetsky and Torrego (2007).

¹⁶Feature interpretation steps cannot be reduced to criterial movement steps in the sense of Rizzi (2006) either. A position is defined as "criterial" if, once a phrase moves to that position, it cannot undergo further movement. However, the literature has shown that extensions of movement chains should not be entirely ruled out, but rather restricted (e.g., Williams 2003; Abels 2007; Neeleman and van de Koot 2010). Formulating such restrictions and deriving them in the grammar is precisely the focus of this paper.

in which movement results in the valuation of several features. In such cases, the lowest-ranked feature serves as the trigger, while the remaining features are valued as free riders. For illustration, consider the hypothetical movement chain of ZP in (50), which involves three distinct movement steps. Suppose that $[\sigma]$ ranks lower than $[\tau]$ on UCOOL. $M_W^{[\sigma]}$ and $M_X^{[\sigma]}$ are triggered by $[\sigma]$, with $[\tau]$ valued as a free rider. The former step targets $[\text{Spec}, \text{WP}]$, whereas the latter targets $[\text{Spec}, \text{XP}]$. The final step, $M_Y^{[\tau]}$, is triggered by $[\tau]$ and targets $[\text{Spec}, \text{YP}]$.



With the notion of *trigger* clarified, we can move on to the mechanism of feature interpretation. Valued features must be interpreted, and each is interpreted once under valuation. In other words, although a feature may trigger multiple movement steps, only a single interpretation step occurs per feature. As a premise, the *height* of probes is defined in terms of asymmetric c-command relations, as in (51). Following Abels (2012b), I assume that a valued feature is interpreted in the highest position in which it served as the trigger, as shown in (52a–b). Conditions in (52c–d) further prevent the interpretation of the same feature elsewhere along the movement path. In (50), the movement chain involves two interpretation steps. While both $M_W^{[\sigma]}$ and $M_X^{[\sigma]}$ are triggered by $[\sigma]$, $M_X^{[\sigma]}$ is the interpretation step for $[\sigma]$, as it targets the highest probe that bears $[\text{u}\sigma]$. Meanwhile, $[\tau]$ is interpreted at the final landing site, $[\text{Spec}, \text{YP}]$.

- (51) An element α in a syntactic object γ is higher than another element β in γ if and only if α asymmetrically c-commands β .
- (52) Principle of interpretation of triggering features:
 A movement step $M_{\beta}^{[F]}$ is the interpretation step for $[F]$ if and only if,
 a. it leads to the valuation of $[F]$ between a head β and its moved specifier, and
 b. movement to a higher projection does not lead to the valuation of $[F]$.
 If so, then
 a. $[F]$ is interpreted on head β ,
 b. for all movement steps $M_{\delta}^{[F]}$ different from $M_{\beta}^{[F]}$, $[F]$ is not interpreted on the corresponding head δ .

The triggering mechanism and the interpretation of composite probes require further clarification. Regarding the triggering mechanism, the cross-linguistic signature of composite A/ $\bar{A}$ constructions provides an important clue. Lohninger (2025) shows that, when a composite $[A/\bar{A}]$ probe finds a fully matching goal, the resulting movement chain is generally an A-chain rather than an $\bar{A}$ -chain or a chain with mixed A- and $\bar{A}$ -properties. Assuming that the triggering feature is the sole determinant of movement-chain properties (van Urk 2015), this cross-linguistic pattern is best explained under UCOOL, where $[A]$ features take precedence over $[\bar{A}]$ features in triggering movement operations (see also Lohninger 2025; for an alternative account, see Cao 2025a). This precedence of $[A]$ over $[\bar{A}]$ features ensures that the resulting movement chain exhibits A-properties even when both features are valued. Therefore, I argue that for composite probes, only the lowest-ranked feature among all bundled features serves as the trigger.

One upshot of this feature-ranking mechanism is that it directly accounts for the A-properties of DP foci versus the $\bar{A}$ -properties of VP foci in the *lian...dou* construction. In Sect. 2, I modeled this state of affairs by positing a composite $[\phi/\text{Foc}]$ probe. DP foci match the $[\phi/\text{Foc}]$ probe in both features, with $[\phi]$ triggering A-movement. In contrast, in cases of VP foci, only $[\text{Foc}]$ is valued, resulting in $\bar{A}$ -movement of VPs. However, a complication I set aside previously is why the valued $[\text{Foc}]$ feature fails to trigger $\bar{A}$ -movement of DP foci? Given that $[\phi]$ belongs to the A-system, whereas $[\text{Foc}]$ belongs to the $\bar{A}$ -system, it should not be disputed that $[\phi]$ ranks lower than $[\text{Foc}]$ ($[\phi] \ll [\text{Foc}]$) in MC. As a result, when both $[\phi]$ and $[\text{Foc}]$ features are valued in cases of DP foci, $[\phi]$ takes precedence in triggering movement, thereby blocking movement triggered by the higher-ranked $[\text{Foc}]$ feature.

Another upshot is that it enables a more fine-grained distinction between bare and composite probes. I argue that the distinction between bare and composite probes lies not in the identification of the trigger but in their interpretation mechanism. Under UCOOL, the trigger in bare and composite probes is identified in a uniform way. When movement results in the valuation of multiple features, whether they act as bare probes or are bundled into composite probes, it is always the lowest-ranked feature—never free riders—that serves as the trigger. In contrast, bare and composite probes differ in how relevant features are interpreted. Previous studies on composite probes argue that features bundled in a composite probe act in unison (e.g., Coon and Bale 2014; van Urk

2015; Bossi and Diercks 2019; Scott 2021; Branan and Erlewine 2024; Lohninger 2025). Under the distinction between feature interpretation and valuation assumed here, I attribute this “act in unison” property to the interpretation of composite probes: all features bundled within a composite probe are interpreted jointly. Specifically, in movement driven by multiple bare probes, each feature is interpreted in its triggering step but never when it is valued as a free rider. In contrast, if movement is driven by composite probes, non-triggering features are also interpreted, even though only the lowest-ranked feature triggers movement. For example, in the *lian...dou* construction, while the $[\phi]$ feature triggers A-movement of DP foci, the non-triggering $[\text{Foc}]$ feature is simultaneously interpreted. I propose a principle of interpretation for composite probes, formulated in (53). It dictates that a composite probe is interpreted at the highest position where its lowest-ranked feature acts as the trigger.

- (53) Principle of interpretation of composite probes:
 For a composite probe $[\sigma/\tau]$, with $\sigma \ll \tau$, all bundled features are interpreted jointly at the interpretation step of $[\sigma]$. The interpretation of $[\sigma]$ follows the principle of interpretation of triggering features in (52).

3.3 Generalized Ban on Improper Feature Interpretation

Abels (2007, 2012b) reaps the effects of UCOOL via the Generalized Prohibition against Improper Movement, as in (54). It imposes the constraint indiscriminately on all movement steps and rules out cases in which movement triggered by a higher-ranked feature is followed by movement triggered by a lower-ranked feature.

- (54) Generalized Prohibition against Improper Movement (GenPIM) (Adapted from Abels 2007, p. 63)
 No constituent may undergo movement triggered by $[\sigma]$ if it has undergone movement of type triggered by $[\tau]$, where $[\sigma] \ll [\tau]$ under UCOOL.

Recall the crucial generalization from (im)proper movement in the *lian...dou* construction: a position that imposes locality constraints on terminal movement into it does not impose the same constraints on intermediate movement into it. Given this generalization, GenPIM must be revised because, by referring to triggering features, it applies uniformly to all movement steps. As I will show in Sect. 4, this formulation is too strong: it would incorrectly rule out well-formed intermediate movements in the *lian...dou* construction.

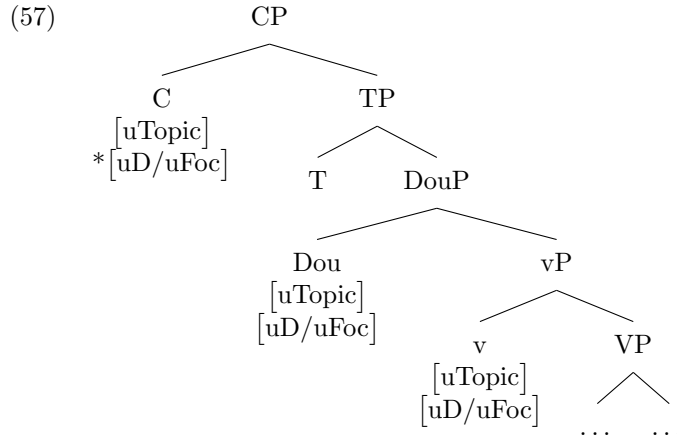
To resolve this problem, I replace GenPIM with a weaker version that applies to *some but not all* movement steps. Specifically, in the current system, terminal movement is recast as feature interpretation steps, a process that is unavailable in intermediate movement steps. On this basis, GenPIM is reformulated as a constraint on possible sequences of feature-interpretation steps, while intermediate movement is ignored, as shown in (55).

- (55) Generalized Ban on Improper Feature Interpretation (GBOIFI)
 No constituent may undergo movement for the interpretation of feature $[\sigma]$ if it has undergone movement for the interpretation of feature $[\tau]$, where $[\sigma] \ll [\tau]$ under UCOOL.

4 Application of the proposal

- (56) (Im)proper cross-clausal movement in the *lian...dou* construction
- | | | | |
|----|------------------------------|------------------------------|--|
| a. | lian-XP ₁ Subject | t' ₁ | dou ... [TP ... t ₁ ... |
| b. | lian-XP ₁ Subject | t' ₁ | dou ... [CP ... [TP ... t ₁ ... |
| c. | Subject lian-XP ₁ | | dou ... [TP ... t ₁ ... |
| d. | * | Subject lian-XP ₁ | dou ... [CP ... [TP ... t ₁ ... |

In this section, I show that once we take the instantiation of UCOOL and the distribution of relevant probes in MC as inputs to the proposed system, the pattern of (im)proper cross-clausal movement in the *lian...dou* construction, as repeated in (56), emerges as a natural consequence. The reconstruction effects we saw in Sect. 2 suggests that topicalization licenses reconstruction for syntactic and semantic relations established by movement driven by $[\phi/\text{Foc}]$. In other words, movement driven by $[\phi/\text{Foc}]$ asymmetrically feeds topicalization. I take it as evidence that in MC, features in the composite $[\phi/\text{Foc}]$ probe rank lower than $[\text{Topic}]$: $[\phi] \ll [\text{Foc}] \ll [\text{Topic}]$.¹⁷ The distinct distribution of $[\text{uD}/\text{uFoc}]$ and $[\text{uTopic}]$ probes is shown in (71). While $[\text{uTopic}]$ probes can be borne by C, v, and *dou* heads, $[\phi/\text{uFoc}]$ probes are present only on *dou* and v heads.¹⁸



¹⁷For a possibly total order of features and associated operations in MC, see Cao (2025b)

¹⁸*Dou* differs from C and v in that its specifier is not an intermediate landing site for topicalization enforced by phase impenetrability. Instead, it serves as an intermediate landing site only when the focus phrase undergoes topicalization. One possible solution is to assume that the presence of $[\text{uTop}]$ on *dou* depends on the presence of $[\phi/\text{uFoc}]$. I believe this solution is within reach in realizational models of morphology, such as Distributed Morphology (e.g., Halle and Marantz 1993; Embick and Noyer 2007) and Nanosyntax (e.g., Starke 2009). Within these frameworks, selectional or combinatorial restrictions can be imposed on syntactic heads, whether these restrictions are syntactic or semantic in nature. These models also provide a natural explanation for the mismatch between the syntactic structure and the morpheme that realizes it. That is, whether or not $[\text{uTop}]$ is present, the head is overtly realized as *dou*. However, I will not pursue this issue here as they concern the syntax-morphology interface (see Cao 2025b for syncretic phenomena involving *dou* and an analysis based on Nanosyntax).

4.1 TP-internal movement

(59) illustrates the derivation of grammatical TP-internal movement, such as example (58). Movement proceeds cyclically through the specifiers of embedded and matrix vPs (59b and e) and terminates at [Spec, DouP] (59g), where the composite $[\phi/\text{Foc}]$ probe is interpreted.

- (58) Li lian $[_{\text{DP}} \text{zhe-ben shu}]_1$ dou bi $[_{\text{TP}} \text{Wu kan-wan le } t_1]$.
 Li lian this-CLF book dou force Wu read-finish ASP
 ‘Li forced Wu to read even this book.’
- (59) a. $[_{v'} v[_{\text{u}\phi/\text{uFoc}}] \dots \text{lian-DP}_{[\phi, \text{Foc}]} \dots$
 b. $[_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$
 c. $[_{\text{TP}} \text{T } [_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$
 d. $[_{v'} v[_{\text{u}\phi/\text{uFoc}}] [_{\text{TP}} \text{T } [_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$
 e. $[_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] [_{\text{TP}} \text{T } [_{\text{vP}} t'_{[\phi/\text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$
 f. $[_{\text{Dou}'} \text{Dou}_{[\text{u}\phi/\text{uFoc}]} [_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] [_{\text{TP}} \text{T } [_{\text{vP}} t'_{[\phi/\text{Foc}]} \dots$
 g. $[_{\text{DouP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{\text{Dou}'} \text{Dou}_{[\phi/\text{Foc}]} [_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$

TP-internal movement cannot cross CPs because C heads do not bear $[\text{u}\phi/\text{uFoc}]$ probes. (61) shows a possible derivation of (60) in which the *lian*-DP moves directly to the higher [Spec, vP] without stopping at the embedded [Spec, CP]. This derivation is ruled out by the PIC. When the higher v is merged, the *lian*-DP becomes trapped in the domain of the embedded C and inaccessible to the $[\text{u}\phi/\text{uFoc}]$ probe on the higher v head. (62) presents an alternative derivation in which the *lian*-DP moves through the embedded [Spec, CP]. Although this derivation obeys the PIC, movement to the embedded [Spec, CP] violates the Last Resort condition.

- (60) *Lisi lian laobao₁ dou jue_{de} $[_{\text{CP}} \text{Zhangsan ma guo } t_1]$
 Lisi lian boss dou think Zhangsan scold EXP
 ‘Intended: ‘Even the boss₁, Lisi thinks Zhangsan scolded him₁.’
- (61) Violations of the Phase Impenetrability Condition
 * $[_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] [_{\text{CP}} \text{C } [_{\text{vP}} t'_{[\phi/\text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$
 ↑
- (62) Violations of Last Resort at CP
 * $[_{\text{vP}} \text{lian-DP}_{[\phi, \text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] [_{\text{CP}} t''_{[\phi/\text{Foc}]} [_{\text{C}'} \text{C } [_{\text{vP}} t'_{[\phi/\text{Foc}]} [_{v'} v[_{\phi/\text{Foc}}] \dots$
 ↑

4.2 TP-external movement

Example (63) shows TP-external movement out of a TP. Its derivation, as depicted in (64), begins with movement driven by $[\phi/\text{Foc}]$, followed by movement triggered by [Topic]. In (64a–f), cyclic movement is driven by $[\phi/\text{Foc}]$, with [Topic] valued as a free rider. Subsequent movement in (64g–h) is driven by [Topic]. $[\phi/\text{Foc}]$ is interpreted in DouP in the matrix clause, since Dou is the highest head bearing $[\phi/\text{Foc}]$. When movement terminates in the matrix CP, the higher-ranked feature [Top] is interpreted there,

as shown in (64h). Given that the order in which $[\phi/\text{Foc}]$ and $[\text{Topic}]$ are interpreted conforms to GBOIFI, the derivation in (64) is well-formed.

- (63) Lian $[\text{DP zhe-ben shu}]_1$ Li dou bi $[\text{TP Wu kan-wan le } t_1]$.
lian this-CLF book Li dou force Wu read-finish ASP
‘Li forced Wu to read even this book.’
- (64) a. $[\text{v}' \text{ v}_{[\text{u}\phi/\text{uFoc}], [\text{uTopic}]}] \dots \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} \dots$
b. $[\text{vP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{v}' \text{ v}_{[\phi/\text{Foc}], [\text{Topic}]}] \dots$
c. $[\text{v}' \text{ v}_{[\text{u}\phi/\text{uFoc}], [\text{uTopic}]}] [\text{TP T} [\text{vP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{v}' \text{ v}_{[\phi/\text{Foc}], [\text{Topic}]}] \dots$
d. $[\text{vP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{v}' \text{ v}_{[\phi/\text{Foc}], [\text{Topic}]}] [\text{TP T} [\text{vP } t'_{[\phi/\text{Foc}], [\text{Topic}]}] \dots$
e. $[\text{Dou}' \text{ Dou}_{[\text{u}\phi/\text{uFoc}], [\text{uTopic}]}] [\text{vP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{v}' \text{ v}_{[\phi/\text{Foc}], [\text{uTopic}]}] [\text{TP} \dots$
f. $[\text{DouP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{Dou}' \text{ Dou}_{[\phi/\text{Foc}], [\text{Topic}]}] [\text{vP } t''_{[\phi, \text{Foc}, \text{Topic}]}] \dots$
g. $[\text{C}' \text{ C}_{[\text{uTopic}]}] [\text{DouP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{Dou}' \text{ Dou}_{[\phi/\text{Foc}], [\text{Topic}]}] [\text{vP} \dots$
h. $[\text{CP lian-DP}_{[\phi, \text{Foc}, \text{Topic}]}] [\text{C}' \text{ C}_{[\text{Topic}]}] [\text{DouP } t'_{[\text{Topic}]}] [\text{Dou}' \dots$

Let us now turn to long-distance TP-external movement, as illustrated in example (65). A puzzle we have seen previously arises from the contrast between ungrammatical TP-internal movement in (66a) and grammatical TP-external movement in (66b). On the one hand, movement triggered by $[\phi/\text{Foc}]$ cannot proceed out of CPs (66a). In the previous section, I attribute this locality restriction to the absence of $[\phi/\text{Foc}]$ on C heads. On the other hand, a $[\text{u}\phi/\text{uFoc}]$ probe located outside the embedded CP can indeed be interpreted. The configuration in (66b) shows that the *lian*-phrase can move out of the embedded CP and interpret $[\phi/\text{Foc}]$ at $[\text{Spec}, \text{DouP}]$ (t'_1) only if it continues moving to the left periphery.

- (65) Lian laobao₁ Lisi t'_1 dou juede $[\text{CP Zhangsan ma guo } t_1]$
lian boss Lisi dou think Zhangsan scold EXP
‘Even the boss₁, Lisi thinks Zhangsan scolded him₁.’
- (66) a. * Subject lian-XP₁ dou ... $[\text{CP} \dots [\text{TP} \dots t_1 \dots$
b. lian-XP₁ Subject t'_1 dou ... $[\text{CP} \dots [\text{TP} \dots t_1 \dots$

To account for the contrast in (66), I argue that TP-external movement in (66b) involves a particular movement path in which the interpretation of $[\phi/\text{Foc}]$ is parasitic on successive-cyclic movement driven by $[\text{Topic}]$. Specifically, $[\text{Spec}, \text{DouP}]$, where $[\phi/\text{Foc}]$ is interpreted, also serves as an intermediate landing site for long-distance topicalization. In what follows, I will show that such movement paths are possible only under the proposed GBOIFI, which constrains the order in which features are interpreted within a movement chain.

The configuration in (67a) shows the derivation within the embedded domain. The *lian*-DP moves cyclically to the embedded $[\text{Spec}, \text{CP}]$, with each step triggered by $[\text{Topic}]$. In (67c), the *lian*-phrase moves from the embedded $[\text{Spec}, \text{CP}]$ to the matrix $[\text{Spec}, \text{vP}]$, resulting in the valuation of both $[\phi/\text{Foc}]$ and $[\text{Topic}]$. Given that $[\phi/\text{Foc}]$ ranks lower than $[\text{Top}]$, this $[\text{Spec}, \text{CP}]$ -to- $[\text{Spec}, \text{vP}]$ movement is triggered by $[\phi/\text{Foc}]$ —more precisely, by $[\phi]$ in the case of DPs and by $[\text{Foc}]$ in the case of VPs.

- (67) a. $[\text{CP } \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} [\text{C}' \text{ C}_{[\text{Topic}]} [\text{vP } \text{t}'_{[\text{Topic}]} [\text{v}' \text{ v}_{[\text{Topic}]}] \dots$
 b. $[\text{v}' \text{ v}_{[\text{u}\phi/\text{uFoc}], [\text{uTopic}]} [\text{CP } \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} [\text{C}' \text{ C}_{[\text{Topic}]}] \dots$
 c. $[\text{vP } \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} [\text{v}' \text{ v}_{[\phi/\text{Foc}], [\text{Topic}]}] [\text{CP } \text{t}''_{[\text{Topic}]} [\text{C}' \text{ C}_{[\text{Topic}]}] \dots$

Notably, it is this intermediate movement step that motivates GBOIFI, proposed as an further refinement of Abels' (2007) GenPIM. Recall that GenPIM applies to all movement steps. Consequently, it undergenerates grammatical TP-external movement in (66b). First, the $[\text{Spec}, \text{CP}]$ -to- $[\text{Spec}, \text{vP}]$ intermediate movement in (67c) is incorrectly predicted to be impossible because GenPIM dictates that movement triggered by $[\text{Topic}]$ cannot be followed by movement triggered by $[\phi/\text{Foc}]$. Second, because of phase impenetrability, there is no alternative way to move a *lian*-phrase from inside the embedded CP to $[\text{Spec}, \text{DouP}]$ in the matrix clause.

In contrast, GBOIFI applies to *some*, *but not all*, movement steps. It constrains possible sequences of feature interpretation steps but ignores feature valuation steps. In (67a–c), although the derivation involves movement triggered by $[\text{Topic}]$ followed by movement triggered by the lower-ranked $[\phi/\text{Foc}]$, it is well-formed under GBOIFI. Movement to the embedded $[\text{Spec}, \text{CP}]$ and the subsequent step to the matrix $[\text{Spec}, \text{vP}]$ are merely feature-valuation steps for $[\text{Top}]$ and $[\phi/\text{Foc}]$, respectively, and are therefore exempt from GBOIFI.

(68) illustrates the subsequent derivation. In (68a), movement to $[\text{Spec}, \text{DouP}]$ results in the valuation of $[\text{D}/\text{Foc}]$ and $[\text{Topic}]$. This movement step is the interpretation step for $[\phi/\text{Foc}]$ and, at the same time, as a feature-valuation step for $[\text{Top}]$. However, at this stage, GBOIFI is not yet invoked; it becomes operative only after all relevant features have been interpreted. As shown in (68c), when the *lian*-DP reaches the matrix $[\text{Spec}, \text{CP}]$, $[\text{Topic}]$ is interpreted at that position. Since the movement step interpreting $[\phi/\text{Foc}]$ is followed by the step interpreting $[\text{Topic}]$, this movement path is permitted by GBOIFI.

- (68) a. $[\text{DouP } \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} [\text{Dou}' \text{ Dou}_{[\phi/\text{Foc}], [\text{Topic}]}] [\text{vP } \text{t}'_{[\phi, \text{Foc}, \text{Topic}]} [\text{v}' \dots$
 b. $[\text{C}' \text{ C}_{[\text{uTopic}]} [\text{DouP } \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} [\text{Dou}' \text{ Dou}_{[\phi/\text{Foc}], [\text{Topic}]}] [\text{vP } \dots$
 c. $[\text{CP } \text{lian-DP}_{[\phi, \text{Foc}, \text{Topic}]} [\text{C}' \text{ C}_{[\text{Topic}]}] [\text{DouP } \text{t}'_{[\text{Topic}]} [\text{Dou}' \dots$

In sum, in MC, $[\phi/\text{Foc}]$ cannot trigger movement across CP boundaries, whereas topicalization can. However, this does not mean that a matrix $[\phi/\text{Foc}]$ probe can never be valued by a goal inside CPs. In cases of long-distance TP-external movement, successive-cyclic topicalization makes available an intermediate landing site in the matrix clause at which $[\phi/\text{Foc}]$ can be interpreted, even though that position would otherwise be inaccessible to movement triggered by $[\phi/\text{Foc}]$ itself.

5 Cyclic movement and clause-internal operations

In this section, I show that the Mandarin pattern discussed above instantiates a more general type of movement path attested cross-linguistically, thereby providing further

support for the proposal advanced here. The languages in Table 1 converge on three properties that jointly define the type of movement path at issue.¹⁹

Table 1 Interactions between movement and local operations

Language	Movement operation	Local operation
Mandarin Chinese	Topicalization	Focus interpretation
Hungarian	Focus/ <i>wh</i> -mvt	Case assignment
Koryak	<i>Wh</i> -mvt	Case assignment
Kiribati	Focus mvt	Object agreement on verbs
Standard Indonesian	<i>Wh</i> -mvt, relativization	Voice morphology on verbs

The first two properties are the relative ordering and locality profiles of syntactic operations, both of which are well established in the literature (e.g., Abels 2007; Neeleman and van de Koot 2010). In each language, the local operation in the final column ranks lower than the movement operation in the third column in the ordering of operations. In previous sections, I showed on the basis of reconstruction effects that, in MC, focus interpretation induced by $[\phi/\text{Foc}]$ ranks lower than topicalization. In the other four languages, the local operations are ϕ -related, whereas the movement operations are different types of $\bar{A}$ -movement. It should not be disputed that these local operations uniformly rank lower than the movement operations. Meanwhile, the movement operations and the local operations exhibit locality mismatches with respect to CPs. The movement operations can extract constituents out of CPs. By contrast, the local operations cannot establish syntactic dependencies across CP boundaries.

The third property is the core movement-path generalization that the present proposal is designed to derive. Namely, the higher-ranked movement operations and the lower-ranked local operations interact in a particular way. A CP-embedded constituent may nonetheless be targeted by the local operation once it undergoes long-distance movement triggered by the movement operation. Notably, neither the movement operation nor the local operation is restricted to a single type. This suggests that such interactions are not attributable to any particular trigger, but instead reflects general properties of long-distance movement: (i) long-distance movement proceeds cyclically through clause-internal intermediate landing sites, and (ii) such intermediate landing sites may be accessible to a lower-ranked operation, provided that these positions fall within the local domain in which that operation can apply.

The interaction between movement and case marking in Koryak, recently identified in Abramovitz (2021), provides a clear instance. Koryak displays patterns of ergative-absolutive case-assignment, with the calculus of case following the algorithm in (69) (a version tailored for the Koryak data; for a more general formulation of case calculus in dependent case theory, see Poole 2023 and references therein).

(69) The calculus of dependent case in Koryak

¹⁹Data from languages other than Mandarin are drawn from the literature: Hungarian (den Dikken 2009a,b; Gervain 2009; János 2014; János et al. 2014), Koryak (Abramovitz 2021), Kiribati (Sabel 2013), Standard Indonesian Sato (2012).

- a. Assign idiosyncratic lexical and inherent cases.
- b. Take the remaining DPs. If DP_α c-commands DP_β in the same local domain, assign ergative case to DP_α .
- c. If a DP was not assigned a case in the previous two steps, then assign it unmarked case (absolutive).

For readability, I omit the glosses of verbal morphology. The sole argument of intransitive verbs must bear absolutive case, as in example (70).²⁰ Some transitive verbs allow more than one case frame, such as *peŋŋ*- ‘attack’. In example (71), when the verb assigns its object dative case—a lexical case in Koryak—the subject receives the unmarked absolutive case. Example (72) shows the case frame of *peŋŋ*- ‘attack’ that does not involve lexical case. In this case, the subject is marked ergative and the object absolutive. Crucially, as shown in (73), these two case frames are in complementary distribution. A dative object and an ergative subject never co-occur.

- (70) (Abramovitz 2021, p. 107)
 {ʔətʔə-n / *ʔətʔ-a} Øviʔi
 {dog-ABS.SG / *dog-ERG} die
 ‘The dog died.’
- (71) (Abramovitz 2021, p. 107)
 kajŋə-n Øpeŋŋe ʔəlva-ŋ
 bear-ABS.SG attack wild.reindeer-DAT
 ‘The bear attacked the wild reindeer.’
- (72) (Abramovitz 2021, p. 107)
 kajŋ-a Øpeŋŋənen ʔəlva-ʔəl
 bear-ERG attack wild.reindeer-ABS.SG
 ‘The bear attacked the wild reindeer.’
- (73) (Abramovitz 2021, p. 107)
 *kajŋ-a Øpeŋŋənen ʔəlva-ŋ
 bear-ERG attack wild.reindeer-DAT
 ‘The bear attacked the wild reindeer.’

Clausal complement-taking verbs behave like intransitive verbs in case assignment. As shown in example (74), the matrix subject is assigned absolutive because it is the only DP in the matrix clause.

- (74) (Abramovitz 2021, p. 120)
 ʔəmmo təvalomək əno ʔewngəto-na-k Øjətʃimawnin kojŋ-o
 1SG.ABS hear that Hewngyto-OBL.SG-ERG break cup-ABL.PL
 ‘I heard that Hewngyto broke cups.’

With case-marking in Koryak clarified for our purpose, we can now turn to the interaction between long-distance movement and local case assignment. Abramovitz (2021) notes that the case marking of the matrix subject is sensitive to long-distance

²⁰In Koryak, there is no split between unaccusatives and unergatives in the case-marking of their subjects.

wh-movement crossing it. As shown in example (75), if the embedded (absolutive) object undergoes *wh*-movement to the left periphery, the matrix subject must bear ergative case, rather than the default absolutive case.

- (75) (Abramovitz 2021, p. 121)
 jej-u₁ {yə-nan / *yətçtç} Øvalomnaw əno ?ewŋəto-na-k
 what-ABS.PL {2SG-ERG / *2SG.ABS} hear that Hewngyto-OBL.SG-ERG
 Øjətçimawnin t₁
 break
 ‘What all did you hear that Hewngyto broke?’

Abramovitz (2021) shows that example (75) indeed involves long-distance *wh*-movement out of the embedded CP, rather than local movement of a proleptic *wh*-phrase within the matrix clause. A proleptic argument and its interpretively related empty category in the embedded clause (presumably a *pro*) are independent arguments.²¹ Consequently, a prolepsis analysis predicts that the case frames of the matrix and the embedded verbs need not be identical, as nothing requires the matrix *wh*-phrase and the embedded *pro* to bear the same case. In other words, an absolutive *wh*-phrase should be able to correspond to a non-absolutive embedded *pro*. Abramovitz (2021) tests this prediction with the examples in (76). In example (76a), the verb *ŋə?* ‘want for’ assigns dative to its object. In the long-distance question in (76b), which targets the embedded object of this verb, the *wh*-phrase cannot be marked absolutive. This contrast shows that a mismatch between the case assigned to the *wh*-phrase and the case assigned to the gap is not allowed. The prediction of the prolepsis analysis is therefore not borne out. By contrast, this pattern is exactly what we expect if the fronted *wh*-phrase and the gap in the embedded clause form a single movement chain, in which the $\bar{A}$ -moved *wh*-phrase preserves the syntactic relations established at its base position.²²

- (76) a. qoɟalqot Økoŋə?ə?Ø petçyə-ŋ.
 Qoɟalqot.ABS.SG want.for food-DAT
 ‘Qoɟalqot wants for food.’
 b. *jənnə ?ewjavana-k Øvalommen yəmmo təkəŋə?əŋ
 what.ABS.SG Hewjava-ERG hear 1SG.ABS want.for
 Intended: ‘What did Hewjava hear that I want for?’

²¹Abramovitz (2021) shows that Koryak allows extensive *pro*-drop.

²²One might expect evidence that, when a dative *wh*-phrase moves to the left periphery, the matrix subject retains its absolutive case. Unfortunately, Abramovitz (2021) reports that extraction from a finite clause is limited to absolutives. As illustrated by the movement of the dative *wh*-phrase in (i), extraction of a non-absolutive is ungrammatical, regardless of the case marking on the matrix subject. Nevertheless, non-absolutives can be extracted from infinitives, and there, as predicted by the calculus of dependent case, they do not trigger ergative case on matrix subjects. For details, see chapter 4 of Abramovitz (2021).

- (i) a. *mekna-ŋ₁ yətçtçi Øvalome əno ?ewŋəto-na-ŋ Øjəlnin urvaq t₁.
 who-DAT 2SG.ABS hear that Hewngyto-OBL.SG-DAT give shirt.ABS.SG
 Intended: ‘Who did you hear that Hewngyto gave a shirt to?’
 b. *mekna-ŋ₁ yə-nan Øvalomən əno ?ewŋəto-na-ŋ Øjəlnin urvaq t₁.
 who-DAT 2SG-ERG hear that Hewngyto-OBL.SG-DAT give shirt.ABS.SG
 Intended: ‘Who did you hear that Hewngyto gave a shirt to?’

Another key fact is that the moved absolutive *wh*-phrase triggers ergative case on the otherwise-absolutive subject at an intermediate landing site within the matrix clause, rather than at the edge of the embedded CP. In each sentence in (77), the embedded clause is an indirect question in which an absolutive *wh*-phrase moves to the embedded [Spec, CP]. However, the matrix subject must bear absolutive case (77a), rather than ergative case (77b). This contrast indicates that, in Koryak, ergative case assignment applies only if the *wh*-phrase moves into an intermediate position above the embedded CP but below the matrix CP.

(77) (Abramovitz 2021, p. 138)

- a. ɣəmmo təkutçetkejɯŋəŋ $[\text{CP } \text{jeq-qevi-jətçɣ-u} \quad \text{məjəlnew}]$
 1SG.ABS think which-gift-contents-ABS.PL give
 ʔewŋəto-na-ŋ $\text{ənək-eto-ɣəlwəje-ŋ.}$
 Hewngyto-OBL.SG-DAT 3SG.POSS-birth-day-DAT
 ‘I am wondering what gifts I should give Hewngyto for his birthday.’
- b. * ɣəm-nan $\text{təkutçetkejɯŋəŋnew}$ $[\text{CP } \text{jeq-qevi-jətçɣ-u} \quad \text{məjəlnew}]$
 1SG-ERG think which-gift-contents-ABS.PL give
 ʔewŋəto-na-ŋ $\text{ənək-eto-ɣəlwəje-ŋ.}$
 Hewngyto-OBL.SG-DAT 3SG.POSS-birth-day-DAT
 ‘I am wondering what gifts I should give Hewngyto for his birthday.’

Abramovitz (2021) attributes the interaction between *wh*-movement and case marking to the dual role of [Spec, vP] in the derivation. First, [Spec, vP] serves as an intermediate landing site of successive-cyclic movement. Second, this position falls within the relevant domain of ergative case assignment, which extends from C down to [Spec, vP] within the same extended projection. Specifically, in example (75), the absolutive *wh*-phrase moves cyclically through matrix [Spec, vP]. At this position, it triggers dependent ergative on the matrix subject by virtue of being the lower of two caseless DPs in the same case domain. This derivation is schematically represented in (78), with the case domain indicated by a square box.

- (78) $[\text{CP } \text{what}_1 \quad \boxed{[\text{C } \dots \text{you}_{\text{ERG}} \dots [\text{vP } t_1'']}] \quad [\text{v}' \text{ v } [\text{VP } \text{hear } [\text{CP } t_1'] \dots]]$

Taken together, although MC and Koryak involve different types of syntactic operations, both instantiate a general type of movement path that any adequate theory of movement should capture: a moving constituent may license independent operations within the domain through which it passes. The present proposal accounts for this general pattern by positing an asymmetry between terminal and intermediate movement with respect to locality constraints on movement. Specifically, in movement triggered a given syntactic operation, the moving constituent remain accessible at its intermediate positions to lower-ranked operations that could not otherwise apply to it once the movement has terminated.

Before closing this section, it is worth noting that this type of movement path is not available in all languages. In English, long-distance *wh*-movement cannot license

ϕ -agreement on matrix T, as shown in (79). However, the contrast between English-type and MC-type movement paths does not undermine the proposed analysis. As emphasized at the beginning of this section, the availability of such movement paths depends not simply on successive-cyclic movement per se, but on whether the intermediate positions created by a higher-ranked movement operation fall within the local domain of a lower-ranked operation.

(79) * Who₁ does t'₁ seem [CP t₁ has left]?

I argue that the ungrammaticality of (79) follows from the fact that, in English, [Spec, TP] is not an intermediate landing site for *wh*-movement. Consequently, long-distance *wh*-movement cannot form a continuous chain passing through matrix TP. Under the present proposal, this is implemented by assuming that T heads do not bear [uWh] probes. (80) illustrates one possible derivation of (79), in which the *wh*-phrase moves from the matrix [Spec, vP] to the matrix [Spec, TP]. Since matrix T does not bear an [uWh] probe, [Wh] is interpreted on the matrix v. After *who* moves to matrix [Spec, TP], [ϕ] on the matrix T is interpreted. Under GBOIFI, this movement path is illicit because [ϕ] cannot be interpreted after [Wh] has already been interpreted. Given that TP is not a phase, an alternative derivation that gives rise to the same surface order is that *wh*-movement proceeds directly from matrix [Spec, vP] to matrix [Spec, CP], as in (81). In this case, however, the [ϕ] probe on T remains unvalued, resulting in an illicit object at the interface.

(80) Violations of the GBOIFI

*[TP Who_{[\phi],[Wh]} [T' T_[\phi] [vP t'_[\phi] [v' v_[Wh] [CP t'_[Wh] [C' C_[Wh] ...

(81) Unvalued features at the interface

*[CP Who_{[\phi],[Wh]} [C' C_[Wh] [TP T_[u\phi] [vP t'_[Wh] [v' v_[Wh] [CP t'_[Wh] [C' C_[Wh] ...

6 The Height-Locality Connection

So far, we have concentrated on the asymmetry between terminal and intermediate movement with respect to locality constraints and, to account for it, I have developed a feature-based formulation of operation ordering. In this section, we turn to an important generalization that previous studies have unearthed regarding the position that hosts terminal movement (for a recent overview, see Keine 2020). I show that the present feature-based account not only preserves the core insight of this generalization, but also carries key implications for it.

We have seen that, in the *lian...dou* construction, TP-external movement is capable of crossing both CPs and TPs, whereas TP-internal movement is restricted to proceed only out of TPs. This pattern corroborates the previous findings that there appears to be a systematic connection between the locality of a movement type (short vs. long) and the structural height of the position that this movement type targets (low vs. high), as stated in (82).

(82) Height-Locality Connection (HLC) (adapted from Keine 2020, p. 12)

The higher the landing site of a movement type is in *fseq*; the more kinds of structures over which this movement type can apply.

The Williams Cycle, as a domain-opacity analysis, captures the HCL by arguing that the height of a movement’s final landing site fully determines its locality profile. It predicts that movement of the general shape in (83) is impossible. That is, a movement type that targets YP cannot move through or cross XP if XP is higher than YP in *fseq*.

$$(83) \quad [_{XP} X \dots [_{YP} \alpha \underbrace{[_{Y'} Y \dots [_{XP} X \dots [_{YP} t_\alpha [_{Y'} Y \dots$$

$$(84) \quad * [_{TP} \text{John}_1 \text{ seems } [_{CP} t_1 \text{ has left}]]$$

Although the Williams Cycle correctly rules out illegitimate hyperraising in English, as in (84), movement constructions that instantiate the configuration in (83) are indeed attested (see Williams 2003; Abels 2007, 2012a; Müller 2014a; Keine 2020; among others). One particular case is legitimate hyperraising (see Zyman 2023 for a recent overview). For example, in the Bantu language Lubukusu, A-movement can cross CP boundaries. In (85a), the embedded verb bears the same noun class marker ‘2’ as the embedded subject *babaandu*, but the matrix verb is marked with ‘6’. This indicates that an in-situ subject agrees only with embedded T. In contrast, in (85b), both of the verbs are assigned the marker ‘2’, due to movement of *babaandu*. That the moved subject controls matrix verb agreement indicates that it is A-movement.

(85) (Carstens and Diercks 2013, p. 100)

- a. $[_{TP} \text{Ka-lolekhana } [_{CP} \text{mbo babaandu ba-kwa }]]$
 6SA-seem that 2people 2SA.PST-fall
 ‘It seems that the people fell.’
- b. $[_{TP} \text{babaandu}_1 \text{ ba-lolekhana } [_{CP} \text{mbo } t_1 \text{ ba-kwa }]]$
 2people 2SA-seem that 2SA.PST-fall
 ‘The people seem like they fell.’

A similar cross-linguistic variation arises in movement targeting the middle field of a clause, commonly referred to as scrambling. In the *lian...dou* construction, TP-internal movement cannot proceed out of CPs (86). By contrast, in Russian, scrambling can proceed out of CPs, as in (87).

- (86) * $Lisi \text{ lian } laobao_1 \text{ dou } juede [_{CP} \text{Zhangsan ma } guo t_1]$
 Lisi lian boss dou think Zhangsan scold EXP
 Intended: ‘Even the boss₁, Lisi thinks Zhangsan scolded him₁.’

- (87) (Keine 2020, p. 195)
 On $[\text{ètu knigu }]_1 \text{ dumajet } [_{CP} \text{čto Pjot pročital } t_1]$
 he this book thinks that Pjot read
 ‘He thinks that Pjotr read this book.’

These data suggest that, while a connection exists between the locality of a movement type and the structural height of its landing site, this connection cannot be reduced to a strict one-to-one mapping between the two. In other words, whether a movement type obeys the HLC is subject to cross-linguistic variation. I do not attempt to provide a comprehensive account of the HLC and its purported exceptions, as this

lies beyond the scope of this paper. Nevertheless, I demonstrate that once the locality of a movement type is attributed to the distribution of its triggering features (as implemented throughout this paper), this cross-linguistic variation emerges as a direct prediction of the present proposal. In short, languages may vary in the distribution of movement-type-specific features, and the locality profiles of these movement types are predicted to vary accordingly (for alternative approaches, see Obata and Epstein 2011; Müller 2014a; Keine 2019; Poole 2023). In this respect, the present proposal contrasts with the Williams Cycle, which is overly restrictive in that it universally prohibits movement dependencies that fall outside the HLC.

For expository purposes, I use the term *altitude* of a feature to denote the highest head in an extended projection that hosts that feature. This notion allows us to state an explicit mapping of feature distribution across extended projections.

In movement that conforms to the HLC, the altitude of the triggering feature is matched across extended projections. To see the consequence of this altitude-matching pattern, suppose that a feature $[\tau]$ distributes over C and v.²³ In other words, $[\tau]$ has C-level altitude across extended projections, as indicated by the horizontal arrow ‘ $\rightarrow$ ’ in the first line of (88). Under this configuration, movement triggered by $[\tau]$ generates long-distance movement targeting a high landing site (e.g., *wh*-movement in English). By contrast, consider now low-altitude cases where $[\tau]$ is confined to v or T, as in (89–90). Movements in (89–90) are clause-bounded and target a low landing site (e.g., TP-internal movement in the *lian...dou* construction and A-movement in English).

- (88) The altitude of $[\tau]$: $C \rightarrow C$
 $[CP \ \alpha_{[\tau]} \ [C' \ C_{[\tau]} \ [vP \ t'''_{[\tau]} \ [v' \ v_{[\tau]} \ [CP \ t''_{[\tau]} \ [C' \ C_{[\tau]} \ [vP \ t'_{[\tau]} \ [v' \ v_{[\tau]} \ \dots$
- (89) The altitude of $[\tau]$: $v \rightarrow v$
a. $[CP \ C \ [vP \ \alpha_{[\tau]} \ [v' \ v_{[\tau]} \ [VP \ t_{[\tau]} \ \dots$
b. * $[CP \ C \ [vP \ \alpha_{[\tau]} \ [v' \ v_{[\tau]} \ [CP \ C \ [vP \ t'_{[\tau]} \ [v' \ v_{[\tau]} \ \dots$
- (90) The altitude of $[\tau]$: $T \rightarrow T$
a. $[CP \ C \ [TP \ \alpha_{[\tau]} \ [T' \ T_{[\tau]} \ [vP \ t_{[\tau]} \ \dots$
b. * $[CP \ C \ [TP \ \alpha_{[\tau]} \ [T' \ T_{[\tau]} \ [CP \ C \ [TP \ t'_{[\tau]} \ [v' \ T_{[\tau]} \ \dots$

Exceptions to the HLC arise from an altitude-mismatching pattern. That is, the altitude of a given feature in one extended projection does not align with its altitude in other extended projections. In (91) and (92), the altitude of $[\tau]$ decreases from C to T and v, respectively, as indicated by the downward arrow ‘ $\searrow$ ’.²⁴ Consequently,

²³The analysis does not depend on the specific identity of $[\tau]$. Rather, we focus on the general shape of movement generated by features of different altitudes.

²⁴Likewise, the altitude of $[\tau]$ may decrease from T to v, as in (i). One may take the English ECM construction as an instance of this configuration, under the analysis in which the embedded subject moves from the embedded TP to a low position in the matrix clause (Postal 1974). This analysis is motivated by the possibility of matrix adverbs intervening between the embedded subject and predicate (e.g., *sincerely* in (ii)). However, on the basis of scope relations and adverb ordering, Neeleman and Payne (2020) argue that ECM infinitives do not involve movement of the embedded subject (ii), but rather rightward partial extraposition of the embedded clause (iii). Since the relevant data instantiating this pattern are largely absent from the literature, I set this issue aside in the main text.

- (i) The altitude of $[\tau]$: $T \searrow v$
 $[TP \ T \ [vP \ \alpha_{[\tau]} \ [v' \ v_{[\tau]} \ [TP \ t'_{[\tau]} \ [T_{[\tau]} \ [vP \ t'_{[\tau]} \ [v' \ v_{[\tau]} \ \dots$
- (ii) John believed Mary₁ (sincerely) $[TP \ t_1 \text{ to be the winner}]$

long-distance movement targets TP and vP domains (e.g., hyperraising in Lubukusu and scrambling in Russian).²⁵

- (91) The altitude of $[\tau]$: $C \searrow T$
 $[_{CP} C [_{TP} \alpha_{[\tau]} [_{T'} T_{[\tau]} [_{vP} t''_{[\tau]} [_{v'} v_{[\tau]} [_{CP} t''_{[\tau]} [_{C'} C_{[\tau]} [_{vP} t'_{[\tau]} [_{v'} v_{[\tau]} \dots$
- (92) The altitude of $[\tau]$: $C \searrow v$
 $[_{CP} C [_{vP} \alpha_{[\tau]} [_{v'} v_{[\tau]} [_{CP} t''_{[\tau]} [C_{[\tau]} [_{vP} t'_{[\tau]} [_{v'} v_{[\tau]} \dots$

Notably, even if feature distribution is arbitrary, the present system still constrains the resulting derivations. In particular, features must be distributed in a way that their interpretation strictly respects their relative ordering. In other words, although any feature could in principle be assigned any altitude, certain combinations are excluded because they inevitably yield improper movement chains. Suppose that there is another feature $[\sigma]$, with $[\sigma]$ ranking lower than $[\tau]$. The present system dictates that, within any extended projection, the altitude of $[\sigma]$ cannot exceed that of $[\tau]$. Example (93) illustrates a case in which $[\sigma]$ has C-level altitude while $[\tau]$ has T-level altitude. In the resulting movement chain, the interpretation step for $[\tau]$ is followed by that for $[\sigma]$, a sequence that is illicit under GBOIFI.

- (93) $* \dots [_{CP} \alpha_{[\sigma], [\tau]} [_{C'} C_{[\sigma]} [_{TP} t'_{[\sigma]} [_{T'} T_{[\tau]} \dots$

Cross-linguistically, movement conforming to the HLC represents the general pattern. From the perspective of language acquisition, the cross-linguistic robustness of the HLC follows naturally if altitude matching is the default setting. By contrast, altitude mismatching can be acquired only on the basis of positive evidence that departs from this default. Children are born assuming the altitude-matching pattern, in the absence of countervailing evidence. For example, if a feature's altitude is C, the child assumes that it maintains this altitude across extended projections. This yields HLC-obeying movement, such as *wh*-movement in English. Alternatively, the learner may encounter evidence that a feature has C-level altitude in one extended projection but T-level altitude in the higher one. Based on such evidence, the default altitude-matching setting is overwritten by an altitude-mismatching one. The child then acquires HLC-violating movement, such as hyperraising in Lubukusu.²⁶

7 Concluding remarks

The point of departure is to adjudicate between domain-opacity and operation-ordering accounts of locality mismatches by exploiting data where the two diverge on the availability of clause-internal intermediate landing sites. I take up this task by investigating cross-clausal movement in the *lian...dou* construction in Mandarin

(iii) John believed $[_{TP} \text{Mary } t_1]$ (sincerely) $[_{to \text{ be the winner}}]_1$

²⁵A third logical possibility is that the altitude of $[\tau]$ increases from one extended projection to another: $v \nearrow T/C$ and $T \nearrow C$. This kind of configuration yields movement targeting higher domains only if the two categories are not separated by another phase head. As far as I can see, this pattern concerns successive cyclicity rather than improper movement, and I therefore leave it aside for reasons of space.

²⁶One remaining question is whether features are associated with a default altitude. Throughout this paper, I maintain the view that the altitude of features is arbitrary and propose a model that does not constrain the altitude itself. As noted by Abels (2007), UCOOL strikingly mirrors the hierarchy of *fseq* targeted by different movement types. I do not know whether this resemblance is coincidental or reflects an underlying principle linking the two, and I leave this issue for future research.

Chinese, where a crucial asymmetry between terminal and intermediate movement is observed. I showed that TP-internal movement in the *lian...dou* construction is clause-bound, whereas TP-external movement can cross CPs. Crucially, the distinction between licit and illicit cross-CP movement hinges on whether TP-internal movement is intermediate or terminal, suggesting that intermediate movement steps must be alleviated from constraints on admissible movement chains. Based on a principled distinction between feature valuation (driving all steps) and feature interpretation (restricted to terminal steps), the longstanding ban on improper movement is reformulated as an order in which triggering features are interpreted along the movement path. It dictates that a movement path is deemed improper if movement that interprets a given feature is extended by movement that interprets a lower-ranked feature. On a general level, the proposed analysis reinforces the feature-based approach to movement (e.g., Abels 2012a,b; van Urk 2015). Interaction between successive-cyclic movement and clause-internal operations, as well as variations in the Height–Locality connection, emerges naturally once features are taken to regulate locality.

In the remainder of this section, I briefly discuss two parallel locality-mismatch effects that fall outside the scope of the present proposal. Although I cannot offer a unified account here, I show that these effects do not contradict the core claim of this paper. Instead, they represent key dimensions that a general theory of locality must ultimately account for, a topic that I leave for future work.

Locality mismatches in non-identity cases Interactions between movement types arise not only when one and the same constituent undergoes two movements (identity cases), but also when distinct constituents that stand in specific structural relations undergo such movements (non-identity cases), as in sub-extraction and remnant movement (e.g., Williams 2003; Abels 2007; Neeleman and van de Koot 2010). For example, a remnant created by A-movement can undergo $\bar{A}$ -movement (94a), but a remnant created by $\bar{A}$ -movement cannot undergo A-movement (94b).

(94) (Keine 2020, p. 307)

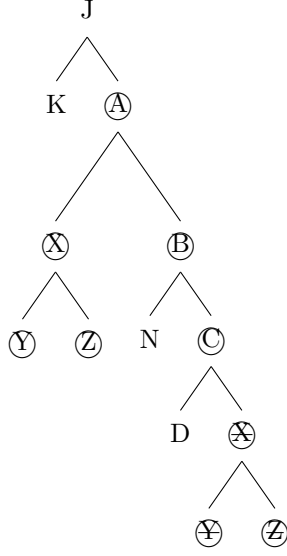
- a. [How likely t_1 to win]₂ is Sue₁ t_2 ?
- b. * [A picture of t_1]₂ is known [which king₁ to have been sold t_2]

These non-identity cases amount to the generalization that a movement type not only renders the moved constituent inaccessible to certain other movement types but also makes certain syntactic domains opaque to them. This generalization is captured by two different implementations of operation ordering. One such implementation is proposed by Abels (2007). Under this view, movement of a constituent α would “affect” not only α itself, but also the elements dominated by α and all nodes along the movement path. These affected nodes, indicated by circles in (95), cannot be targeted by movements ranked lower than the affecting movement.

A different implementation is proposed by Neeleman and van de Koot (2010). They argue that the notion of *affectedness* is stipulated and address this issue by proposing an alternative interpretation of movement. Within this framework, the trace of movement introduces a selectional requirement (SR) that percolates up the tree and triggers reconstruction once it is satisfied. This mechanism ensures that the circled

nodes in (95) are precisely those that cannot be targeted by a lower-ranked movement, as doing so would block the reconstruction of SR to its trace.

(95) (Abels 2007, p. 13)



Crucially, the core insight of the present proposal—namely, the asymmetry between terminal and intermediate movement with respect to locality constraints—is compatible with both implementations. In Abels (2007), specifiers along the movement path (N and D in (95)), which serve as intermediate landing sites for successive-cyclic movement, are never affected. Since neither the locality constraint proposed here nor the notion of *affectedness* refers to intermediate movement, the latter can be incorporated into the present proposal. Under Neeleman and van de Koot’s (2010) implementation, what requires revision is the constraint on reconstruction, according to which SR_1 reconstructs for SR_2 only if SR_1 ranks higher than SR_2 . I contend that such a reformulation is also feasible, given that the exemption of intermediate movement from locality constraints is already implied in their analysis. Specifically, under their analysis, reconstruction is triggered only when SR reaches its licensing position; intermediate positions therefore remain outside the scope of the relevant constraint. Since neither implementation directly regulates intermediate movement, both can naturally accommodate the asymmetry proposed here, making a unified theory that encompasses identity and non-identity cases readily available.

Locality mismatches in non-movement dependencies Recent studies have shown that non-movement dependencies exhibit locality-mismatch effects analogous to those found in movement dependencies (Keine 2019, 2020; Poole 2023). For illustration, consider long-distance agreement (LDA) in Hindi. In example (96), gender agreement on both the matrix and embedded verbs are obligatorily controlled by *film* within the embedded TP. Masculine singular default agreement is not allowed. In contrast, in (97), the matrix verb cannot agree with *ghazal* inside the embedded CP and must instead display default agreement, even if *ghazal* is located at the edge of

the embedded CP. This contrast indicates that LDA exhibit analogous locality mismatches: it can established non-movement dependencies across TP boundaries, but is blocked across CP boundaries.

- (96) (Keine 2019, p. 21)
- | | | | | | | | | |
|--------------------------|-----------------------|---|------------------------|--------------|---|-----------------|----------------|---------|
| har | bacce-ko ₁ | [| us-kii ₁ | maa-ne |] | [_{TP} | t ₁ | film |
| every | child-DAT | | 3SG-GEN | mother-ERG | | | | movie.F |
| dikhaa-nii/*-naa | | | | caah-ii/*-aa | | | | |
| show-INF.F.SG/*-INF.M.SG | | | want-PF.F.SG/*-PF.M.SG | | | | | |
- ‘For every child *x*, *x*’s mother wanted to show *x* a movie.’
- (97) (Keine 2019, p. 25)
- | | | | | | | |
|-------------|-------------------------|---|-----------------|----------|----------|--------------|
| firoz-ne | soc-aa/*-ii | | [_{CP} | ghazal | monaa-ne | gaa-yii |
| FirozERG | think-PF.M.SG/*-PF.F.SG | | | ghazal.F | Mona-ERG | sing-PF.F.SG |
| th-ii | |] | | | | |
| be.PST-F.SG | | | | | | |
- ‘Firoz thought that Mona had sung ghazal.’

The studies cited above generally adopt a domain-opacity approach. To account for the contrast between (96) and (97), Keine (2019) posits that a probe is associated with a *horizon* that determines which domains it can search into. In Hindi, the ϕ -probe is located on T and cannot search for a goal inside domains higher in *fseq*. As discussed in the Introduction, however, this approach incorrectly rules out clause-internal intermediate landing sites in successive-cyclic movement. A unified theory of movement and non-movement dependencies, if such a theory is possible, should explain the properties of both dependency types without sacrificing independently motivated properties of either. Therefore, I argue that a domain-opacity analysis cannot by itself provide a satisfactory account of the parallel effects found in movement and non-movement dependencies.

However, this does not mean that a general theory is unattainable. This paper contributes to the study of locality mismatches by clarifying the core analytical intuition that such a general theory should capture. Only by disentangling intermediate movement from the relevant locality constraint can we see the true analogy between movement and non-movement dependencies. Specifically, locality systematically rules out certain combinations of two linked positions. In movement dependencies, these linked positions are the launching site and the final landing site; in non-movement dependencies, they are the positions of the licensor and the licensee.

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